

450 Notes

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Definition 0.1 [Exterior Measure]. Given any set $\emptyset \neq X \subset \mathbb{R}^n$, an exterior/outer measure μ^* on X is a function

$$\mu^* : \mathcal{B}(X) \rightarrow [0, \infty]$$

by

$$\mu^*(X) = \inf \sum_{i=1}^{\infty} (b_i - a_i)$$

where $\mathcal{B}(X)$ is the power set of X and $X \subset \bigcup_{i=1}^{\infty} [a_i, b_i]$ which satisfies

1. $\mu^*(\emptyset) = 0$
2. (monotone increasing) If $E \subset F \subset X$, then $\mu^*E \leq \mu^*F$ and
3. (countable subadditivity) $\mu^*(\bigcup_{i=1}^{\infty} A_i) \leq \sum_{i=1}^{\infty} \mu^*(A_i)$

Definition 0.2 [Translation Invariant]. We say an exterior measure μ^* is translation invariant if for all $E \subset \mathbb{R}$ and all $c \in \mathbb{R}$, we have

$$\mu^*(E + c) = \mu^*(E)$$

where $E + c := \{x + c : x \in E\}$

0.1 Properties of Exterior Measure

1. Let $E \subset \mathbb{R}^n$ be a countable set. Then $\mu^*E = 0$
2. Exterior measure is translation invariant.
3. Exterior does not have additive property.

Proof. 1. Since E is countable, we can write $E = \{x_n\}_{n=1}^{\infty}$. Let $\epsilon > 0$, and for each $n \geq 1$, set $I_n = (x_n - \frac{\epsilon}{2^{n+1}}, x_n + \frac{\epsilon}{2^{n+1}})$. Then $\{I_n\}_{n=1}^{\infty}$ is a cover of E . Thus,

$$0 \leq \mu^*(E) \leq \sum_{n=1}^{\infty} l(I_n) = \sum_{n=1}^{\infty} \frac{\epsilon}{2^n} = \epsilon$$

Since $\epsilon > 0$ is arbitrary, we have $\mu^*(E) = 0$

2. Observe that $\{I_n\}_{n=1}^\infty$ is a cover of E (by open intervals) if and only if $\{I_n + \kappa\}_{n=1}^\infty$ is a cover of $E + \kappa$ (by open intervals). Moreover, for any interval $I = (a, b)$ with $a < b$, we have

$$\ell(I) = b - a = (b + \kappa) - (a + \kappa) = \ell(I + \kappa),$$

while if I is of the form $(-\infty, b)$, (a, ∞) , or $(-\infty, \infty)$, then $\ell(I) = \infty = \ell(I + \kappa)$. Thus

$$\begin{aligned} m^*E &= \inf \left\{ \sum_{n=1}^\infty \ell(I_n) : \{I_n\}_{n=1}^\infty \text{ a cover of } E \text{ by open intervals} \right\} \\ &= \inf \left\{ \sum_{n=1}^\infty \ell(I_n + \kappa) : \{I_n + \kappa\}_{n=1}^\infty \text{ a cover of } E + \kappa \text{ by open intervals} \right\} \\ &= m^*(E + \kappa). \end{aligned}$$

3. See A1Q3. □

1 Lebesgue Measure

After introducing the exterior measure, we now have the tool to "measure". And here's the new question: what does it mean for a set to be measurable? What are some special properties about this kind of set?

1.1 Definition of measurable and Caratheodory condition

We shall prove that the definition and C condition are equivalent:

Definition 1.1 [Lebesgue measure and measurable]. A subset E of $\mathbb{R}^d$ is Lebesgue measurable, or simply measurable, if for any $\epsilon > 0$ there exist an open set O with $E \subset O$ and

$$\mu^*(O - E) \leq \epsilon$$

If E is measurable, then we define its **Lebesgue measure** μ to be the restriction of μ^* to $\mathfrak{M}(\mathbb{R})$, where $\mathfrak{M}(\mathbb{R})$ denote the set of all measurable sets. That is, Lebesgue measure is the function

$$\mu : \mathfrak{M}(\mathbb{R}) \rightarrow [0, \infty]$$

$$E \mapsto \mu^*(E)$$

In this case, $\mu(E) = \mu^*(E)$

Definition 1.2 [Caratheodory condition]. Let $E \subset \mathbb{R}^n$, then E is measurable iff for all $A \subset \mathbb{R}^n$,

$$\mu^*(A) = \mu^*(A \cap E) + \mu^*(A \cap E^c)$$

Proof. Since $A = (A \cap E) \cup (A \cap E^c)$, we have $\mu^*(A) \leq \mu^*(A \cap E) + \mu^*(A \cap E^c)$ by property of exterior measure. So we only need to prove the other direction.

($\implies$) Suppose E is measurable. WLOG, suppose $\mu^*(E) < \infty$. Let $\epsilon > 0$, there exist open set O_E with $E \subset O_E$ such that $\mu^*(O_E - E) \leq \epsilon$. Note

$$\begin{aligned} \mu^*(A) &\leq \mu^*(A \cap E) + \mu^*(A \cap E^c) \\ &\leq \mu^*(A \cap O_E) + \mu^*((A \cap O_E^c) \cup (A \cap E^c \cap O_E)) \\ &\leq \mu^*(A \cap O_E) + \mu^*(A \cap O_E^c) + \mu^*(O_E - E) \\ &\leq \mu^*(A) + \epsilon \end{aligned}$$

Let $\epsilon \rightarrow 0$, then " $=$ " holds.

($\impliedby$) Suppose the Caratheodory condition holds. WLOG, assume $\mu^*(E) < \infty$. For all $\epsilon > 0$, there exist open set U_ϵ with $E \subset U_\epsilon$ and $\mu^*(U_\epsilon) < \mu^*(E) + \epsilon$. By the condition, we have

$$\begin{aligned} \mu^*(U_\epsilon) &= \mu^*(U_\epsilon \cap E) + \mu^*(U_\epsilon \cap E^c) \\ &= \mu^*(E) + \mu^*(U_\epsilon - E) \end{aligned}$$

$$\implies \mu^*(U_\epsilon - E) = \mu^*(U_\epsilon) - \mu^*(E) < \epsilon$$

Hence, E is measurable by definition. $\square$

1.2 Ideas to prove a set is measurable

1. Use the definition of measurable sets to express the set in a way such that it can be covered by some open sets.
2. Use the C condition together with set computations (i.e. De Morgan's Law). To prove equality, we can always give ourselves some space to prove the inequality involve some $\epsilon > 0$, then let $\epsilon \rightarrow 0$.
3. If we are given some measurable sets, try to find the connection of the set we're trying to prove and the measurable set and use monotonicity of measure to get a range of measures (i.e. the non-measurable set $\mathcal{N}$).
4. Use the fact that every open and every closed set is measurable and the properties of σ -algebra.
5. To prove infinite arguments, first prove it under finite cases, say for some N the argument holds true, then let $N \rightarrow \infty$ to get a conclusion.

Definition 1.3 [σ -algebra]. Let X be a set. A σ -algebra on X is a collection $\mathcal{B}$ of X which obeys the following properties:

1. (Empty set) $\emptyset \in \mathcal{B}$.
2. (Complement) If $E \in \mathcal{B}$, then the complement $E^c := X \setminus E$ also lies in $\mathcal{B}$.
3. (Countable unions) If $E_1, E_2, \dots \in \mathcal{B}$, then $\bigcup_{n=1}^{\infty} E_n \in \mathcal{B}$.

A Borel σ -algebra in $\mathbb{R}^d$, denoted by $\mathbb{B}_{\mathbb{R}^d}$, which by definition is the **smallest σ -algebra that contains all open sets**. Elements of this σ -algebra are called **Borel Sets**.

We refer to the pair $(X, \mathcal{B})$ of a set X together with a σ -algebra on that set as a **measurable space**.

Lemma 1.4 (Exterior measure 0 implies measurable). *If $\mu^*(E) = 0$, then E is measurable. In particular, if F is a subset of a set of exterior measure 0, then F is measurable.*

Proof. For every $\epsilon > 0$ there exists an open set $\mathcal{O}$ with $E \subset \mathcal{O}$ and $\mu^*(\mathcal{O}) \leq \epsilon$. Since $(\mathcal{O} \setminus E) \subset \mathcal{O}$, monotonicity implies $\mu^*(\mathcal{O} \setminus E) \leq \epsilon$, as desired.

As a consequence of this property, we deduce that the Cantor set $\mathcal{C}$ is measurable and has measure 0. $\square$

Lemma 1.5 (limit of measurable set). *1. If $E_k \uparrow E$, then $E \in \mathcal{B}_{\mathbb{R}}$ with*

$$\mu(E) = \lim_{k \rightarrow \infty} \mu(E_k).$$

2. If $E_k \downarrow E$ and $m(E_1) < \infty$, then

$$\mu(E) = \lim_{k \rightarrow \infty} \mu(E_k).$$

Proof. 1. By definition,

$$E = \bigcup_{k=1}^{\infty} E_k = \biguplus_{k=1}^{\infty} (E_k \setminus E_{k-1}), \quad (E_0 = \emptyset).$$

Hence,

$$\begin{aligned} \mu(E) &= \sum_{k=1}^{\infty} \mu(E_k \setminus E_{k-1}) = \lim_{N \rightarrow \infty} \sum_{k=1}^N \mu(E_k \setminus E_{k-1}) \\ &= \lim_{N \rightarrow \infty} \mu\left(\bigcup_{k=1}^N (E_k \setminus E_{k-1})\right) = \lim_{N \rightarrow \infty} \mu(E_N). \end{aligned}$$

2. Note

$$E_1 = E \cup \bigcup_{k=0}^{\infty} (E_k \setminus E_{k+1}), \quad (\text{disjoint unions}).$$

Thus,

$$\begin{aligned} \mu(E_1) &= \mu(E) + \sum_{k=0}^{\infty} \mu(E_k \setminus E_{k+1}) \\ &= \mu(E) + \lim_{N \rightarrow \infty} \sum_{k=0}^N \mu(E_k \setminus E_{k+1}) \end{aligned}$$

$$\begin{aligned}
&= \mu(E) + \lim_{N \rightarrow \infty} \mu \left(\bigcup_{k=0}^N (E_k \setminus E_{k+1}) \right) \\
&= \mu(E) + \lim_{N \rightarrow \infty} (\mu(E_0) - \mu(E_N)).
\end{aligned}$$

Since $\mu(E_0) < \infty$,

$$\mu(E) = \lim_{N \rightarrow \infty} \mu(E_N).$$

Note: Statement (2) is false without the assumption $\mu(E_1) < \infty$.

For example, $(n, \infty) \downarrow \emptyset$, but

$$\mu(n, \infty) = \infty \quad \text{while} \quad \mu(\emptyset) = 0.$$

□

1.3 Example

For the non-measurable set $\mathcal{N}$ we constructed in class, any measurable set of $\mathcal{N}$ has measure 0. The proof is in A3Q1.

However, the general version of this argument: any measurable subset of a non-measurable set has measure 0 is NOT TRUE. Consider the set $A = \mathcal{N} \cup [3, 4]$. Then A is the non-measurable. Indeed, if A is measurable, then $\mathcal{N} = A^c \cap [3, 4]$ is measurable, by σ -algebra properties. This is a contradiction. Thus, A is non-measurable. However, $[3, 4] \subset A$ which has measure 1.

1.4 Cantor set

Recall the construction of Cantor set. It turns out that any Cantor-like set (i.e. base n without some i , $1 \leq i \leq n$) inherit the same properties from Cantor set. That is, they are measurable with measure 0.

When construct counter examples, it is often helpful to consider Cantor set as well as the vitali's set.

Example 1.6 [Cantor-Lebesgue Function]. Consider the function $f : \mathcal{C} \rightarrow [0, 1]$ which maps 3 base number to base 2 representation. That is,

$$2 * 0.b_1 b_2 \mapsto 0.b_1 b_2$$

where $b_i \in \{0, 1\}$

Proof.

Claim: $f \uparrow$:

$$\text{Suppose } 2 \sum_{n=1}^{\infty} \frac{\alpha_n}{3^n} < 2 \sum_{n=1}^{\infty} \frac{\beta_n}{3^n}.$$

Let n_0 be the smallest n so that $\alpha_n \neq \beta_n$. Claim $\alpha_{n_0} = 0$, $\beta_{n_0} = 1$.

$$2 \left(\frac{\beta_{n_0}}{3^{n_0}} - \frac{\alpha_{n_0}}{3^{n_0}} \right) + 2 \sum_{n > n_0} \frac{2}{3^n} > 0$$

$$\frac{\beta_{n_0}}{3^{n_0}} - \frac{\alpha_{n_0}}{3^{n_0}} + \frac{1}{3^{n_0}} > 0$$

$$\Rightarrow \beta_{n_0} - \alpha_{n_0} + 1 > 0$$

$$\Rightarrow \alpha_{n_0} = 0, \quad \beta_{n_0} = 1.$$

$$\Rightarrow f \left(2 \sum_{n=1}^{\infty} \frac{\alpha_n}{3^n} \right) = \sum_{n=1}^{\infty} \frac{\alpha_n}{2^n}$$

$$\leq \sum_{n=1}^{\infty} \frac{\beta_n}{2^n} = f \left(2 \sum_{n=1}^{\infty} \frac{\beta_n}{3^n} \right).$$

□

Definition 1.7 [Cantor Lebesgue Function]. The Cantor-Lebesgue function $F : \mathcal{C} \rightarrow [0, 1]$ is defined on $\mathcal{C}$ by

$$F(x) = \sum_{k=1}^{\infty} \frac{b_k}{2^k} \quad \text{if} \quad x = \sum_{k=1}^{\infty} a_k 3^{-k}, \quad \text{where } b_k = \frac{a_k}{2}.$$

where $a_k = 0, 2$.

Equivalently, The Cantor-Lebesgue function $F : \mathcal{C} \rightarrow [0, 1]$ takes every removed interval of the Cantor set to be horizontal lines and connect the other parts by positive slope lines. Figure 1 below shows an example:

1.5 Construction of CL

Consider the standard triadic Cantor set $\mathcal{C} \subset [0, 1]$ described at the end of Section 1 in Chapter 1, and recall that

$$\mathcal{C} = \bigcap_{k=0}^{\infty} C_k,$$

where each C_k is a disjoint union of 2^k closed intervals. For example,

$$C_1 = [0, 1/3] \cup [2/3, 1].$$

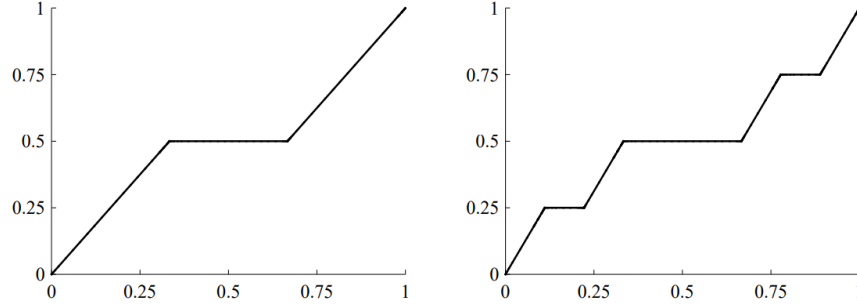


Figure 1: Cantor-Lebesgue Function

Let $F_1(x)$ be the continuous increasing function on $[0, 1]$ that satisfies $F_1(0) = 0$, $F_1(x) = 1/2$ if $1/3 \leq x \leq 2/3$, $F_1(1) = 1$, and F_1 is linear on C_1 . Similarly, let $F_2(x)$ be continuous and increasing, and such that

$$F_2(x) = \begin{cases} 0, & \text{if } x = 0, \\ 1/4, & \text{if } 1/9 \leq x \leq 2/9, \\ 1/2, & \text{if } 1/3 \leq x \leq 2/3, \\ 3/4, & \text{if } 7/9 \leq x \leq 8/9, \\ 1, & \text{if } x = 1, \end{cases}$$

and F_2 is linear on C_2 . This process yields a sequence of continuous increasing functions $\{F_n\}_{n=1}^{\infty}$ such that

$$|F_{n+1}(x) - F_n(x)| \leq 2^{-n-1}$$

Let $F = \lim_{n \rightarrow \infty} F_n$, then $\{F_n\}_{n=1}^{\infty} \rightarrow F$ uniformly. Hence, F is continuous. Note F is constant on each interval of the complement of the Cantor set. Since $m(C) = 0$, we have $F'(x) = 0$ a.e.

1.6 Properties of Cantor-Lebesgue Function

1. CL is continuous.
2. CL is surjective.
3. $F'(x) = 0$ a.e.

Proof. The other two properties are proved in the construction section, so we only prove surjective in this case:

Fix any $y \in [0, 1]$. Choose a binary expansion

$$y = \sum_{k=1}^{\infty} \frac{b_k}{2^k}, \quad b_k \in \{0, 1\},$$

using the convention that if y has two binary expansions (i.e. y is dyadic), we take the one that does not end in infinitely many 1's (equivalently, the terminating one).

Define ternary digits

$$a_k := 2b_k \in \{0, 2\}, \quad k \geq 1,$$

and set

$$x := \sum_{k=1}^{\infty} a_k 3^{-k}.$$

Since each $a_k \in \{0, 2\}$, we have $x \in C$. By definition of F on C , the associated binary digits of x are exactly $b_k = a_k/2$, hence

$$F(x) = \sum_{k=1}^{\infty} \frac{b_k}{2^k} = y.$$

Because $y \in [0, 1]$ was arbitrary, $F(C) = [0, 1]$, i.e. F is surjective. $\square$

1.7 Remark

Think about what the graph looks like when a function is given.

2 Measure Space and completion

Definition 2.1 [Lebesgue–Stieltjes]. The **Lebesgue–Stieltjes** integral

$$\int_a^b f(x) dg(x)$$

is defined when $f : [a, b] \rightarrow \mathbb{R}$ is Borel-measurable and bounded and $g : [a, b] \rightarrow \mathbb{R}$ is of bounded variation in $[a, b]$ and right-continuous, or when f is non-negative and g is monotone non-decreasing and right-continuous.

Definition 2.2 [Measurable Space]. We refer to the pair (X, Σ, μ) of a set X as a **measurable space**, where Σ denote the set of σ -measure and μ is the measure on X .

Definition 2.3 [Completion of Measurable Space]. A set $E \subset X$ is a **null set** for a measure μ on X if $E \in \Sigma$ with $\mu(E) = 0$, where Σ denote the σ -algebra.

A **complete measure** is one such that every subset A of every null set E is measurable.

We say (X, Σ_0, μ_0) is the **completion** of (X, Σ, μ) if Σ_0 is the smallest σ -algebra containing the original one and μ_0 is the complete measure of μ .

In other words, completion of measurable space adds all "missing" subsets of measure-zero sets to the σ -algebra

Example 2.4. For a measurable space $(\mathbb{R}, \mathcal{B}_{\mathbb{R}}, \mu_f)$ satisfying

$$\mu_f([a, b]) = f(b) - f(a)$$

2.1 Find the completion

The key idea is to find the null set and add these sets to the new σ -algebra.

For example, find the completion of $(\mathbb{R}, \mathcal{B}_{\mathbb{R}}, \mu_f)$ where $f(x) = e^{2x}$.

Proof. By definition, $\mu_f((a, b]) = f(b) - f(a) = e^{2b} - e^{2a}$. Next, we consider the null set. If $\mu_f(E) = 0$, then by Lebesgue–Stieltjes, $\mu_f(E) = \int_E f'(x) d\mu(x) = 0$. Hence, $\int_E 2e^{2x} d\mu(x) = 0$. Since $\frac{d\mu_f}{d\mu} = 2e^{2x} > 0$, $\mu(E) = 0$. Conversely, if $\mu(E) = 0$, then $\int_E 2e^{2x} d\mu(x) = 0 = \mu_f(E)$. Thus, μ_f and μ have the same null set. Therefore, the completion is itself. $\square$

Definition 2.5 [Dirac Measure]. A *Dirac measure* is a measure δ_x on a set X (with any σ -algebra of subsets of X) defined for a given $x \in X$ and any (measurable) set $A \subseteq X$ by

$$\delta_x(A) = \mathbf{1}_A(x) = \begin{cases} 0, & x \notin A, \\ 1, & x \in A. \end{cases}$$

where $\mathbf{1}_A$ is the indicator function of A .

Example 2.6 $[\mu_f]$. If $f(x) = e^{2x}$, then

$$\begin{aligned} \mu_f([a, b]) &= f(b) - f(a) \\ &= e^{2b} - e^{2a} \\ &= \int_a^b 2e^{2x} dx. \end{aligned}$$

Example 2.7 [completion].

$$\textbf{Claim: } \overline{(\mathbb{R}, \mathcal{B}(\mathbb{R}), \delta_0)} = (\mathbb{R}, 2^{\mathbb{R}}, \delta_0).$$

Recall the completion of (X, Σ, μ) is $(X, \overline{\Sigma}, \overline{\mu})$, where

$$\overline{\Sigma} = \{A \cup N : A \in \Sigma, \exists Z \in \Sigma \text{ with } \mu(Z) = 0 \text{ and } N \subseteq Z\},$$

and $\overline{\mu}(A \cup N) = \mu(A)$ (well-defined).

Let $\mu = \delta_0$ on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$. Then for any set $E \subseteq \mathbb{R}$,

$$\delta_0(E) = \mathbf{1}_E(0) = \begin{cases} 1, & 0 \in E, \\ 0, & 0 \notin E. \end{cases}$$

Hence a Borel set Z is δ_0 -null iff $0 \notin Z$. In particular,

$$\mathbb{R} \setminus \{0\} \in \mathcal{B}(\mathbb{R}) \quad \text{and} \quad \delta_0(\mathbb{R} \setminus \{0\}) = 0.$$

Therefore, in the completion, every subset $N \subseteq \mathbb{R} \setminus \{0\}$ is measurable.

Now let $A \subseteq \mathbb{R}$ be arbitrary. If $0 \notin A$, then $A \subseteq \mathbb{R} \setminus \{0\}$, so

$$A = \emptyset \cup A \in \overline{\Sigma}.$$

If $0 \in A$, then

$$A = \{0\} \cup (A \setminus \{0\}),$$

where $\{0\} \in \mathcal{B}(\mathbb{R})$ and $A \setminus \{0\} \subseteq \mathbb{R} \setminus \{0\}$, so again

$$A \in \overline{\Sigma}.$$

Thus every subset $A \subseteq \mathbb{R}$ lies in $\overline{\Sigma}$, i.e.

$$\overline{\Sigma} = 2^{\mathbb{R}}.$$

Consequently, the completion of $(\mathbb{R}, \mathcal{B}(\mathbb{R}), \delta_0)$ is

$$(\mathbb{R}, 2^{\mathbb{R}}, \delta_0).$$

3 Measurable functions and Littlewoods' principle

Definition 3.1 [Measurable function]. A function f defined on the **measurable set** $E \subset \mathbb{R}^d$ is said to be **measurable** if $\forall a \in \mathbb{R}$

$$f^{-1}([-\infty, a)) = \{x \in E, f(x) < a\}$$

is measurable.

1. Every measurable set of finite measure is almost a finite union of intervals
2. Every measurable function is almost continuous (Lusin's Theorem).
3. A pointwise convergence sequence of measurable functions is almost uniformly convergent. (Egorov)

3.1 Properties of measurable function

Property 2. If f is continuous on $\mathbb{R}^d$, then f is measurable. If f is measurable and finite-valued, and Φ is continuous, then $\Phi \circ f$ is measurable.

Proof. Suppose f is continuous, then for every open set $\mathcal{O}$, the preimage $f^{-1}(\mathcal{O})$ is also open and thus measurable.

In fact, Φ is continuous, so $\Phi^{-1}((-\infty, a))$ is an open set $\mathcal{O}$, and hence $(\Phi \circ f)^{-1}((-\infty, a)) = f^{-1}(\mathcal{O})$ is measurable.

It should be noted, however, that in general it is *not* true that $f \circ \Phi$ is measurable whenever f is measurable and Φ is continuous. See Exercise 35.

□

Property 3. Suppose $\{f_n\}_{n=1}^{\infty}$ is a sequence of measurable functions. Then

$$\sup_n f_n(x), \quad \inf_n f_n(x), \quad \limsup_{n \rightarrow \infty} f_n(x), \quad \liminf_{n \rightarrow \infty} f_n(x)$$

are measurable.

Proof. Proving that $\sup_n f_n$ is measurable requires noting that

$$\{\sup_n f_n > a\} = \bigcup_n \{f_n > a\}.$$

This also yields the result for $\inf_n f_n(x)$, since this quantity equals $-\sup_n(-f_n(x))$.

The result for the limsup and liminf also follows from the two observations:

$$\limsup_{n \rightarrow \infty} f_n(x) = \inf_k \left\{ \sup_{n \geq k} f_n \right\}, \quad \liminf_{n \rightarrow \infty} f_n(x) = \sup_k \left\{ \inf_{n \geq k} f_n \right\}.$$

□

Property 4. If $\{f_n\}_{n=1}^{\infty}$ is a collection of measurable functions, and

$$\lim_{n \rightarrow \infty} f_n(x) = f(x),$$

then f is measurable.

Proof. Since

$$f(x) = \limsup_{n \rightarrow \infty} f_n(x) = \liminf_{n \rightarrow \infty} f_n(x),$$

this property is a consequence of Property 3.

□

Property 5. If f and g are measurable, then:

- (i) The integer powers f^k , $k \geq 1$, are measurable.
- (ii) $f + g$ and fg are measurable if both f and g are finite-valued.

Lemma 3.2 (Monotone increasing $\implies$ measurable). *If $f : \mathbb{R} \rightarrow \mathbb{R}$ is monotone increasing, then f is Borel measurable (and hence is also Lebesgue measurable).*

Proof. Fix $a \in \mathbb{R}$. Since f is monotone increasing, $\{f > a\}$ is either an interval of the form (x, ∞) , or it is an interval of the form $[x, \infty)$. Each of these are Borel sets, so f is Borel measurable. $\square$

Lemma 3.3 (Continuous $\implies$ measurable). *If $f : \mathbb{R}^d \rightarrow \mathbb{R}$ is continuous, then f is Borel measurable (and hence is also Lebesgue measurable).*

Proof. Fix $a \in \mathbb{R}$. Since the interval (a, ∞) is an open subset of $\mathbb{R}$ and f is continuous, the inverse image $f^{-1}(a, \infty)$ is open in $\mathbb{R}^d$ and hence is a Borel set. Since $\{f > a\} = f^{-1}(a, \infty)$, this shows that f is Borel measurable. $\square$

3.2 Egorov

Suppose $\{f_k\}_{k=1}^\infty$ is a sequence of measurable functions defined on a measurable set E with $m(E) < \infty$, and assume that $f_k \rightarrow f$ a.e. on E . Given $\varepsilon > 0$, we can find a closed set $A_\varepsilon \subset E$ such that

$$m(E - A_\varepsilon) \leq \varepsilon \quad \text{and} \quad f_k \rightarrow f \text{ uniformly on } A_\varepsilon.$$

3.2.1 Remark

What Egorov really states is that if we have pointwise convergence, then after removing a measure zero set, we can extend to uniform convergence!

The key idea to this prove is to construct the set based on the definition of uniform convergence and see what will happen as the set changes. In addition, when deal with an interval that involve non-measurable and measurable sets, one can always split the interval into measurable parts and non-measurable parts.

3.3 Lusin

Suppose f is measurable and finite valued on E with E of finite measure. Then for every $\varepsilon > 0$ there exists a closed set F_ε , with

$$F_\varepsilon \subset E, \quad \text{and} \quad m(E - F_\varepsilon) \leq \varepsilon,$$

and such that $f|_{F_\varepsilon}$ is continuous.

3.3.1 Remark

The key idea of proving Lusin is that for all measurable function f over $[0, 1]$, there exist step function ϕ_n such that $\phi_n \rightarrow f$ a.e.

Proof. Let f_n be a sequence of step functions so that $f_n \rightarrow f$ a.e. Then we may find sets E_n so that $m(E_n) < 1/2^n$ and f_n is continuous outside E_n . By Egorov's theorem, we may find a set $A_{\varepsilon/3}$ on which $f_n \rightarrow f$ uniformly and $m(E - A_{\varepsilon/3}) \leq \varepsilon/3$. Then we consider

$$F' = A_{\varepsilon/3} - \bigcup_{n \geq N} E_n$$

for N so large that $\sum_{n \geq N} 1/2^n < \varepsilon/3$. Now for every $n \geq N$ the function f_n is continuous on F' ; thus $\bar{f}$ (being the uniform limit of $\{f_n\}$) is also continuous on F' . To finish the proof, we merely need to approximate the set F' by a closed set $F_\varepsilon \subset F'$ such that $m(F' - F_\varepsilon) < \varepsilon/3$. $\square$

3.4 simple function approximation theorem

Definition 3.4 [Simple Function]. Let (X, Σ) be a measurable space. A simple function on X is a measurable function $\varphi : X \rightarrow \mathbb{C}$ of the form

$$\varphi(x) = \sum_{i=1}^n a_i \chi_{E_i}$$

where $E_i \cap E_j = \emptyset$ for $i \neq j$. That is, φ takes only finitely many distinct values.

Below figure 2 gives an idea of approximate measurable function with simple functions.

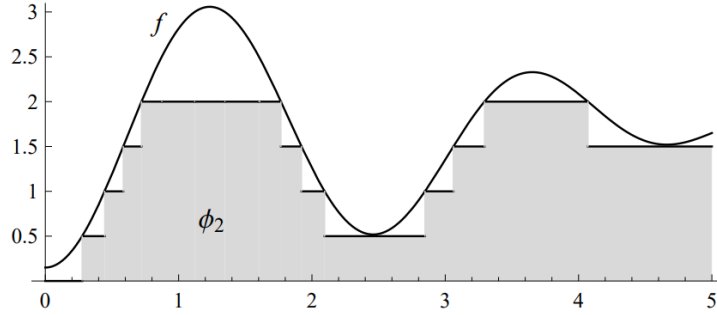


Fig. 4.2. Graphs of a function f and the approximating simple function ϕ_2 (with shading under the graph of ϕ_2).

Figure 2: Simple function approximation

3.5 Note

1. To check a function f is measurable, one can always split f to be $f = f^+ - f^-$. To take the integration of f , one would take the integration of both parts, i.e. $\int f^+ dx = \sup_{\phi} \{ \int \phi dx : 0 \leq \phi \leq f, \phi \text{ is simple} \}$. Notice in this case, f does not need to be measurable, while both f^+ and f^- are measurable.
2. Continuous functions are measurable.

4 Convergence

The key idea is to compute integration using continuous functions, which can be approximated by simple functions (i.e. characteristic functions). To prove uniformly convergence, construct a set that satisfies the convergence condition.

4.1 Dominated Convergence Theorem

Suppose $\{f_n\}$ is a sequence of measurable functions such that $f_n(x) \rightarrow f(x)$ a.e. x , as n tends to infinity. If $|f_n(x)| \leq g(x)$, where g is integrable, then f_n, f are integrable and

$$\lim_{n \rightarrow \infty} \int_S f_n d\mu = \int_S \lim_{n \rightarrow \infty} f_n d\mu = \int_S f d\mu$$
$$\int |f_n - f| \rightarrow 0 \quad \text{as } n \rightarrow \infty,$$

and consequently

$$\int f_n \rightarrow \int f \quad \text{as } n \rightarrow \infty.$$

4.2 Monotone Convergence Theorem

Let $\{f_n\}$ be a sequence of non-negative non-decreasing measurable functions with $f_n \rightarrow f$. Then

$$\lim_{n \rightarrow \infty} \int f_n = \int f$$

4.2.1 Remark

The DCT allows us to interchange the limit and integration under certain condition. Together with MCT, it gives us an idea of how does a function behaviour when it approaches its limit. At the same time, Fatou's Lemma provide more information about the bounded of the function. In all of these cases, the function only need to be measurable instead of continuous.

4.3 Dini's Theorem

If X is a compact topological space, and $(f_n)_{n \in \mathbb{N}}$ is a monotonically increasing sequence (meaning $f_n(x) \leq f_{n+1}(x)$ for all $n \in \mathbb{N}$ and $x \in X$) of continuous real-valued functions on X which converges pointwise to a continuous function $f : X \rightarrow \mathbb{R}$, then the convergence is uniform.

The same conclusion holds if $(f_n)_{n \in \mathbb{N}}$ is monotonically decreasing instead of increasing.

4.4 Fatou's Lemma

Suppose $\{f_n\}$ is a sequence of measurable functions with $f_n \geq 0$. If $\lim_{n \rightarrow \infty} f_n(x) = f(x)$ for a.e. x , then

$$\int f \leq \liminf_{n \rightarrow \infty} \int f_n.$$

Lemma 4.1 (limit = sup). *Limit of a non-decreasing sequence equals its supremum. That is,*

$$f(x) = \liminf_{n \rightarrow \infty} f_n(x) = \sup_n \inf_{k \geq n} f_k(x) = \sup_n g_n(x).$$

5 $L^1(\mathbb{R})$ Space

We care about complete measure space since it allows us to treat sets of measure zero being negligible. The fact that the integrable functions form a vector space is an important observation about the algebraic properties of such functions. A fundamental analytic fact is that this vector space is complete in the appropriate norm.

Definition 5.1 [$L^1(\mathbb{R})$]. $L^1(\mathbb{R}) = \{f \text{ is measurable} : \int_{\mathbb{R}} |f(x)| dx < \infty\}$

$$\|g\|_{\infty} = \inf\{c : \mu(|g| > c) = 0\} < \infty$$

See the next example.

Theorem 5.2 (Riesz-Fischer). *The vector space L^1 is complete in its metric.*

Proof. Suppose (f_n) is a Cauchy sequence, we want to find $f \in L^1(\mathbb{R})$ so that

$$\int |f_n - f| \rightarrow 0 \quad \text{as } n \rightarrow \infty$$

Passing to a subsequence, we can assume

$$\|f_{n_k} - f_{n_{k+1}}\|_1 < 2^{-k}$$

Set

$$f = f_{n_1} + \sum_{k=1}^{\infty} (f_{n_{k+1}} - f_{n_k})$$

$$g = |f_{n_1}| + \sum_{k=1}^{\infty} |f_{n_{k+1}} - f_{n_k}|$$

We know from **MCT** (Monotone Convergence Theorem) that

$$\begin{aligned} \int g &< \int |f_{n_1}| + \sum_{k=1}^{\infty} 2^{-k} < \infty \\ \implies g &< \infty \quad \text{a.e.} \end{aligned}$$

$$\implies f \in L^1(\mathbb{R}),$$

since

$$|f| \leq g$$

By definition:

$$f_{n_k} \rightarrow f \quad \text{a.e.}$$

Since

$$|f_{n_k}| < g \quad \text{and} \quad \int g < \infty$$

By DCT, we have

$$\begin{aligned} \lim_{k \rightarrow \infty} \int |f_{n_k} - f| &= 0 \\ \implies d(f_{n_k}, f) &\rightarrow 0 \\ \implies d(f_n, f) &\rightarrow 0 \end{aligned}$$

□

5.1 Example

Consider

$$\|g\|_\infty = \inf\{c : \mu(|g| > c) = 0\} < \infty$$

This seems confused, however, we can think about the graph. It's a function with 2 pieces: one is "main body" of the function while the other one is just ∞ . Draw any horizontal line $y = c$, it may or may not have the interception with the function itself. Project ∞ down to the x-axis, if there's nothing in between the ∞ and $y = c$, the measure will be 0. Otherwise, $\mu > 0$. $\|g\|_\infty$ is the smallest such c such that it satisfies the condition. We define a bounded linear function Φ , then by its linearity, we have $\|\Phi_g(f)\| \leq \|g\|_\infty \|f\|_1$. It turns out that all bounded linear functions arise in this way.

Definition 5.3 [Signed Measure]. Let $(X, \mathcal{A})$ be a measurable space. A signed measure on $(X, \mathcal{A})$ is a function $\mu : \mathcal{A} \rightarrow \mathbb{R}^*$ such that

1. μ takes on at most one of the values $-\infty$ or ∞ .
2. $\mu(\emptyset) = 0$
3. If $\{E_n\}_{n=1}^\infty \subset \mathcal{A}$ is a sequence of pairwise disjoint sets, then

$$\mu\left(\bigcup_{n=1}^\infty E_n\right) = \sum_{n=1}^\infty \mu(E_n).$$

5.2 Mutually Singular and absolutely continuous

Definition 5.4 [Mutually singular and absolutely continuous]. Two signed measures ν and μ on $(X, \mathcal{M})$ are **mutually singular** if there are disjoint subsets A and B in $\mathcal{M}$ so that

$$\nu(E) = \nu(A \cap E) \quad \text{and} \quad \mu(E) = \mu(B \cap E) \quad \text{for all } E \in \mathcal{M}.$$

Thus ν and μ are supported on disjoint subsets. We use the symbol $\nu \perp \mu$ to denote the fact that the measures are mutually singular.

Equivalently, two signed measures μ, ν on $(X, \mathcal{B})$ are *mutually singular* (written $\mu \perp \nu$) if there is a decomposition

$$X = A \cup B, \quad A \cap B = \emptyset,$$

such that A is ν -null and B is μ -null (i.e. $\nu(A) = 0$ and $\mu(B) = 0$).

Definition 5.5 [absolutely continuous functions]. A function $f : [a, b] \rightarrow \mathbb{R}$ is **absolutely continuous** (AC) if for all $\varepsilon > 0$, there exists a $\delta > 0$ such that whenever (x_i, y_i) are disjoint intervals in $[a, b]$ and $\sum_{i=1}^n (y_i - x_i) < \delta$, then

$$\sum_{i=1}^n |f(y_i) - f(x_i)| < \varepsilon.$$

Definition 5.6 [Absolutely continuous (measure)]. In contrast, if ν is a signed measure and μ a (positive) measure on $\mathcal{M}$, we say that ν is *absolutely continuous* with respect to μ if

$$\nu(E) = 0 \quad \text{whenever } E \in \mathcal{M} \text{ and } \mu(E) = 0. \quad (1)$$

That is, $\mu(E) = 0 \implies \nu(E) = 0$

Thus if ν is supported in a set A , then A must be an essential part of the support of μ in the sense that $\mu(A) > 0$. We use the symbol $\nu \ll \mu$ to indicate that ν is **absolutely continuous** with respect to μ . Note that if ν and μ are mutually singular, and ν is also absolutely continuous with respect to μ , then ν vanishes identically.

5.2.1 Note

- From the definition, it is clear that absolutely continuous functions are continuous, and in fact uniformly continuous.
- If $F(x) = \int_a^x f(y)dy$, where f is integrable, then F is absolutely continuous.
- If F is absolutely continuous on $[a, b]$, then $F'(x)$ exists a.e. Moreover, if $F'(x) = 0$ for a.e. x , then F is constant.

5.3 Radon–Nikodym Theorem

Let (X, Σ) be a measurable space, let μ, ν be two σ -finite measures. If $\nu \ll \mu$, (i.e. $\mu(E) = 0 \implies \nu(E) = 0$) then there exist measurable function

$$f : X \rightarrow [0, \infty]$$

such that for any measurable set $A \in \Sigma$

$$\nu(A) = \int_A f d\mu$$

5.4 Hahn–Jordan Decomposition

Every signed measure μ defined on Σ has a *unique* decomposition into the difference $\mu = \mu^+ - \mu^-$ of two positive measures, μ^+ and μ^- , at least one of which is finite, such that $\mu^+(E) = 0$ for every Σ -measurable subset $E \subseteq N$ and $\mu^-(E) = 0$ for every Σ -measurable subset $E \subseteq P$, for any Hahn decomposition (P, N) of μ . We call μ^+ and μ^- the *positive* and *negative part* of μ , respectively. The pair (μ^+, μ^-) is called a *Jordan decomposition* (or sometimes *Hahn–Jordan decomposition*) of μ . The two measures can be defined as

$$\mu^+(E) := \mu(E \cap P) \quad \text{and} \quad \mu^-(E) := -\mu(E \cap N)$$

for every $E \in \Sigma$ and any Hahn decomposition (P, N) of μ .

Note that the Jordan decomposition is unique, while the Hahn decomposition is only *essentially* unique.

Proof. Consider

$$\mathcal{P} = \{P : \nu|_P \text{ is a measure}\}.$$

Let

$$a = \sup_{P \in \mathcal{P}} \nu(P) < \infty.$$

Then there exist sets $P_n \in \mathcal{P}$ such that

$$\nu(P_n) \uparrow a.$$

Let

$$P^+ = \bigcup_{n=1}^{\infty} P_n.$$

Then $\nu|_{P^+}$ is a measure, and moreover

$$\nu(P^+) = a.$$

Indeed, $\nu_{P^+}(E) = \nu(E \cap P^+) = \lim_{n \rightarrow \infty} \nu(E \cap P_n) = \lim_{n \rightarrow \infty} \nu_{P_n}(E)$. Thus, $\nu|_{P^+}$ is a measure because a pointwise limit of an increasing sequence of measures is itself a measure.

Let P^+ be as before, and define

$$P^- = (P^+)^c.$$

Claim. The set function $-\nu_{P^-}$ is a measure.

Suppose not. Then there exists the smallest $n_1 \in \mathbb{N}$ such that

$$\{P \subseteq P^- : \nu(P) > 1/n_1\} \neq \emptyset.$$

Choose $P_1 \subseteq P^-$ with $\nu(P_1) > 1/n_1$.

Since ν_{P^-} is not a measure, there exists a smallest $n_2 \in \mathbb{N}$ such that

$$\{P \subseteq P_1 : \nu(P) > \nu(P_1) + 1/n_2\} \neq \emptyset.$$

Choose $P_2 \subseteq P_1$ such that

$$\nu(P_2) > \nu(P_1) + 1/n_2.$$

Continuing inductively, we obtain a decreasing sequence

$$P_1 \supseteq P_2 \supseteq P_3 \supseteq \cdots$$

and integers n_k such that

$$\nu(P_{k+1}) > \nu(P_k) + 1/n_{k+1}.$$

Let

$$P = \bigcap_{k=1}^{\infty} P_k.$$

Then

$$\nu(P) = \lim_{k \rightarrow \infty} \nu(P_k) \geq \sum_{k=1}^{\infty} \frac{1}{n_k} = \infty,$$

a contradiction. Hence $-\nu_{P^-}$ is a measure.

Let

$$P = \bigcap_{k=1}^{\infty} P_k.$$

Then

$$a \geq \nu(P) = \lim_{k \rightarrow \infty} \nu(P_k) \geq \sum_{k=1}^{\infty} \frac{1}{n_k}.$$

Since $n_k \rightarrow \infty$ as $k \rightarrow \infty$, the series $\sum_{k=1}^{\infty} \frac{1}{n_k}$ diverges.

Hence there exists $\tilde{P} \subset P$ such that

$$\nu(\tilde{P}) \geq \nu(P) + \frac{1}{n}$$

for some n .

For $n_k > n$ we have $\tilde{P} \subseteq P_k$ for k large, so

$$\nu(\tilde{P}) \geq \nu(P_k) + \frac{1}{n},$$

a contradiction. □

5.5 $(L^1(\mathbb{R}))^* \cong L^\infty(\mathbb{R})$

Definition 5.7 [σ -finite measure]. Let $(X, \mathcal{M})$ be a measurable space and let $\mu : \mathcal{M} \rightarrow [0, \infty]$ be a measure. We say that μ is σ -finite if there exist measurable sets $E_1, E_2, \dots \in \mathcal{M}$ such that

$$X = \bigcup_{n=1}^{\infty} E_n \quad \text{and} \quad \mu(E_n) < \infty \text{ for all } n \in \mathbb{N}.$$

Equivalently, the measure space $(X, \mathcal{M}, \mu)$ is called σ -finite.

5.5.1 Draft Proof

The key ideas are:

1. We need to prove ν is a signed measure and thus a σ -finite measures so that we can use the Lebesgue-Radon-Nikodym Theorem. To prove ν is signed measure, we only need to prove the first two assumptions, since if ν takes both positive and negative infinity, then let $\nu(A) = \infty$, and $\nu(B) = -\infty$, by sub-additivity, $\nu(A) + \nu(B) = \infty + (-\infty)$ which is undefined. Hence, ν can not take both values.

Proof. It is clearly that $\emptyset \neq (L^1(\mathbb{R}))^*$. Let $\phi \in (L^1(\mathbb{R}))^*$. For n fixed, we define

$$v(E) = \phi(\chi_E), E \subset [-n, n]$$

We claim this is a signed measure, here's the proof:

Notice that $v(\emptyset) = \phi(\chi_\emptyset) = \phi(0) = 0$. Let $E = \bigcup_{n=1}^N E_n$

$$v(E) = \phi(\chi_E) = \phi(\chi_{\sum_{n=1}^N E_n}) = \sum_{n=1}^N v(E_n)$$

Let $N \rightarrow \infty$, we have

$$v(\bigcup_{n=1}^{\infty} E_n) = \sum_{n=1}^{\infty} v(E_n)$$

Thus, v is a signed measure.

Also we know
$$\sum_{i=1}^N \chi_{E_i} \xrightarrow{L^1([-n,n])} \sum_{i=1}^{\infty} \chi_{E_i}$$

$$\Rightarrow \lim_{N \rightarrow \infty} \phi \left(\sum_{i=1}^N \chi_{E_i} \right) = \phi \left(\sum_{i=1}^{\infty} \chi_{E_i} \right)$$

$$\lim_{N \rightarrow \infty} \sum_{i=1}^N \nu(E_i) = \nu\left(\bigcup_{i=1}^{\infty} E_i\right)$$

Thus, if

$$\mu(E) = 0 \Rightarrow \chi_E = 0 \text{ in } L^1(\mu) \Rightarrow \nu(\chi_E) = 0 \Rightarrow \nu \ll \mu$$

By Hahn-Jordan Decomposition, we may write $\nu = \nu^+ - \nu^-$, where $\nu^+ \perp \nu^-$ and thus $\nu^+, \nu^- \ll \mu$. WLOG, assume ν is positive. Now if we can find a function g such that $d\nu = g d\mu$, then we can easily show $g \in L^\infty(\mathbb{R})$. To do this, we first construct a set

$$\mathcal{G} = \{g : [-n, n] \rightarrow [0, \infty] \text{ measurable} \mid \int_E g d\mu \leq \nu(E) \forall E \in \mathcal{M}\}$$

Note: if

$$g_1, g_2 \in \mathcal{G}, \text{ then } h = \max\{g_1, g_2\} \in \mathcal{G}.$$

Let $A = \{g_1 > g_2\}$. Then

$$\int_E h d\mu = \int_{E \cap A} h d\mu + \int_{E \cap A^c} h d\mu = \int_{E \cap A} g_1 d\mu + \int_{E \cap A^c} g_2 d\mu \leq \nu(E \cap A) + \nu(E \cap A^c) = \nu(E).$$

Hence $h \in \mathcal{G}$.

Let

$$a = \sup \left\{ \int g d\mu : g \in \mathcal{G} \right\}.$$

Then there exists a sequence $g_k \in \mathcal{G}$ such that

$$\int g_k d\mu \nearrow a.$$

Let

$$\tilde{g}_k = \max(g_1, g_2, \dots, g_k),$$

then

$$\int \tilde{g}_k d\mu \nearrow a \quad \text{and} \quad \tilde{g}_k \nearrow g.$$

By MCT, $\int g d\mu = a$. Now we need to prove that actually $d\nu = g d\mu$:

Note that $d\nu - g d\mu$ defines a measure λ on $([-\eta, \eta], \mathcal{M})$. We want to show it is the zero measure. Consider

$$\lambda_k = \lambda - \frac{1}{k} \mu.$$

Write

$$\lambda_k = \lambda_k^+ - \lambda_k^-,$$

and let $[-\eta, \eta] = P_k^+ \cup P_k^-$ with $P_k^+ \cap P_k^- = \emptyset$.

Define

$$P^+ = \bigcup_k P_k^+, \quad P^- = \bigcap_k P_k^-.$$

Then $(\lambda - \frac{1}{k}\mu)(P^-) \leq 0$ for all k , so

$$\lambda(P^-) = 0.$$

If $\mu(P^+) = 0$, then $\lambda \perp \mu$ since

$$[-\eta, \eta] = P^+ \cup P^-.$$

Otherwise, if $\mu(P^+) > 0$, then $\mu(P_k^+) > 0$ for some k , and $\lambda \geq \frac{1}{k}\mu$ on P_k^+ .

$$\Rightarrow d\nu - g d\mu \geq \frac{1}{k}d\mu \quad \text{on } P_k^+.$$

Thus

$$d\nu \geq g d\mu + \frac{1}{k}d\mu \quad \text{on } P_k^+.$$

Hence

$$\int (g + \frac{1}{k}\chi_{P_k^+}) d\mu > a, \quad \text{and} \quad g + \frac{1}{k}\chi_{P_k^+} \in \mathcal{G}.$$

So we must have

$$\lambda \perp \mu \quad \text{and} \quad \lambda \ll \mu.$$

Hence $\lambda = 0$

□

5.5.2 Remark

The fact that $(L^1(\mathbb{R}))^* \cong L^\infty(\mathbb{R})$ is essential since it gives us an idea of what a continuous linear functional on L^1 looks like. In other words, any bounded linear function Φ on $L^1(\mu)$ is represented by integration against a unique function $g \in L^\infty(\mu)$:

$$\Phi(f) = \int_X fg d\mu \quad \text{for all } f \in L^1(\mu).$$

Similarly, we can show that $(L^1(\mathbb{R}))^{**} = (L^1(\mathbb{R})) = (L^\infty(\mathbb{R}))^*$

6 Fourier Transformation and Fourier Series

6.1 Motivation

(One of the amazing references : <https://www.jezzamon.com/fourier/index.html>)
Consider a case where we are dealing with sounds editing. We know that every sound is just a wave of certain period and frequency (i.e. "Do" is 440 Hz) and each wave has its shape. Now suppose we are given the shape of a curve, we want to know how many tones are there in the shape and what is the composition of that sound?

After Fourier Transform, we can turn the sound wave into a nice, clear wave of separate sound with peaks and each Hz. Now consider a intensity vs time graph. We use a vector to parametrize the path of that curve and draw the position of center of mass in another graph. Since complex plane are good to

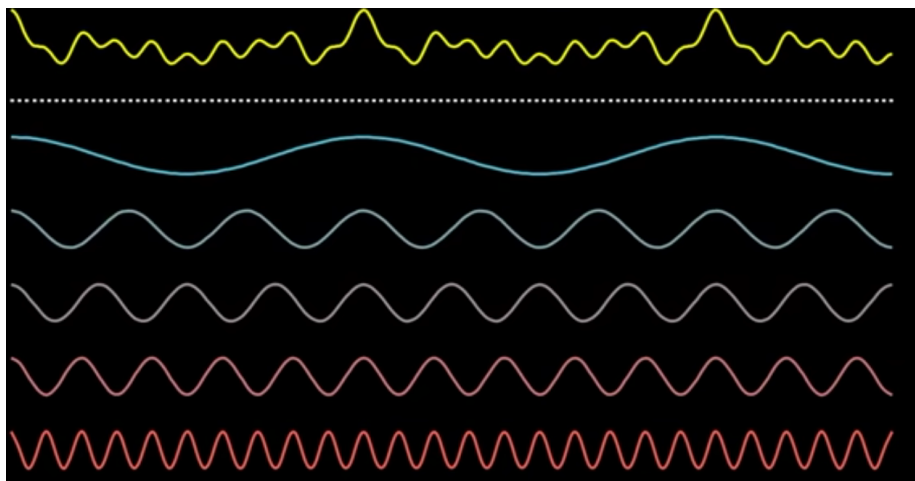


Figure 3: Decomposition of sound

deal with rotation and winding, we will put the set up to be in complex plane. By Euler's formula, if we are given $e^{2\pi i t}$, it will be landed on the complex unit circle. Then we can use $e^{2\pi i t}$ to describe the position with respect to t . This may move too fast to be tracked, so let ξ be the frequency, $e^{2\pi i \xi t}$ describe the same position slower than the previous one.

Now Fourier transform consider clockwise direction. Suppose we are given the original function as $g(t)$, then $g(t)e^{-2\pi i \xi t}$ describe the wind the graph around the circle with length $g(t)$ clock-wisely. If we choose points in the original graph and we track their position, we'll get the approximated shape of the transformed graph. The more points we have, the more accurate we get. Hence, we take the integral of the original graph. That is,

$$\hat{g}(\xi) = \int g(t)e^{-2\pi i \xi t} dt$$

is the Fourier transform.

Definition 6.1 [Fourier Transform]. The Fourier Transform is defined as :

$$\hat{f}(\xi) = \int_{\mathbb{R}^d} f(x)e^{-2\pi i \xi x} dx$$

6.2 Thms

Proposition 6.2. Suppose $f \in L^1(\mathbb{R}^d)$. Then $\hat{f}$ defined above is continuous and bounded on $\mathbb{R}^d$.

Proof. In fact, since $|f(x)e^{-2\pi i x \cdot \xi}| = |f(x)|$, (one can think about the e part as describing the shape of the unit circle.) (Indeed, $|f(x)e^{-2\pi i x \cdot \xi}| = |f(x)||e^{-2\pi i x \cdot \xi}| =$

$|f(x)|$ because they are complex values so $|ab|^2 = (ab)\bar{a}\bar{b} = |a|^2|b|^2 \implies |ab| = |a||b|$) the integral representing $\hat{f}$ converges for each ξ and

$$\sup_{\xi \in \mathbb{R}^d} |\hat{f}(\xi)| \leq \int_{\mathbb{R}^d} |f(x)| dx = \|f\|_1.$$

To verify the continuity, note that for every x , $f(x)e^{-2\pi i x \cdot \xi} \rightarrow f(x)e^{-2\pi i x \cdot \xi_0}$ as $\xi \rightarrow \xi_0$, where ξ_0 is any point in $\mathbb{R}^d$; hence $\hat{f}(\xi) \rightarrow \hat{f}(\xi_0)$ by the dominated convergence theorem.

One can assert a little more than the boundedness of $\hat{f}$; namely, one has $\hat{f}(\xi) \rightarrow 0$ as $|\xi| \rightarrow \infty$, but not much more can be said about the decrease at infinity of $\hat{f}$. (See Exercises 22 and 25.) As a consequence, for general $f \in L^1(\mathbb{R}^d)$ the function $\hat{f}$ is not in $L^1(\mathbb{R}^d)$, and the presumed formula (15) becomes problematic. The following theorem evades this difficulty and yet is useful in a number of situations. $\square$

Lemma 6.3 ($\hat{f} = 0 \implies f = 0$ a.e.). *If $\hat{f} = 0$, then $f = 0$ a.e.*

Proof. Since $f \in L^1(\mathbb{R})$ and $\hat{f} \in L^1(\mathbb{R})$, the Fourier inversion theorem applies and yields

$$f(x) = \int_{\mathbb{R}} \hat{f}(\xi) e^{2\pi i x \xi} d\xi \quad \text{for almost every } x \in \mathbb{R}.$$

By assumption, $\hat{f}(\xi) = 0$ for almost every ξ , hence the integrand $\hat{f}(\xi) e^{2\pi i x \xi}$ is zero for almost every ξ and for every fixed x . Therefore the integral on the right-hand side is equal to 0 for every x , so

$$f(x) = 0 \quad \text{for almost every } x \in \mathbb{R}.$$

This shows that f is the zero function in $L^1(\mathbb{R})$, as claimed. $\square$

Theorem 6.4 (Fubini's Theorem). *Suppose X and Y are σ -finite measure spaces and suppose that $X \times Y$ is given the product measure (which is unique as X and Y are σ -finite). Fubini's theorem states that if f is $X \times Y$ integrable, meaning that f is a measurable function and*

$$\int_{X \times Y} |f(x, y)| d(x, y) < \infty,$$

then

$$\int_X \left(\int_Y f(x, y) dy \right) dx = \int_Y \left(\int_X f(x, y) dx \right) dy = \int_{X \times Y} f(x, y) d(x, y).$$

Theorem 6.5 (Tonelli's Theorem). *Let $(X, \mathcal{A}, \mu)$ and $(Y, \mathcal{B}, \nu)$ be σ -finite measure spaces, and let $f : X \times Y \rightarrow [0, \infty]$ be a non-negative measurable function. Then*

$$\int_{X \times Y} f(x, y) d(\mu \times \nu)(x, y) = \int_X \left(\int_Y f(x, y) d\nu(y) \right) d\mu(x) = \int_Y \left(\int_X f(x, y) d\mu(x) \right) d\nu(y),$$

and all three integrals may take the value $+\infty$. Moreover, for μ -almost every $x \in X$, the function $y \mapsto f(x, y)$ is ν -measurable, and for ν -almost every $y \in Y$, the function $x \mapsto f(x, y)$ is μ -measurable.

Proposition 6.6.

$$\lim_{t \rightarrow 0} \int_{\mathbb{R}} e^{-4\pi^2 \xi^2 t} \widehat{f}(\xi) e^{2\pi i \xi x} d\xi = f(x).$$

Proof.

$$\begin{aligned} \int_{\mathbb{R}} e^{-4\pi^2 \xi^2 t} \widehat{f}(\xi) e^{2\pi i \xi x} d\xi &= \int_{\mathbb{R}} e^{-4\pi^2 \xi^2 t} \left(\int_{\mathbb{R}} f(y) e^{-2\pi i y \xi} dy \right) e^{2\pi i \xi x} d\xi \\ &= \int_{\mathbb{R}} f(y) \left(\int_{\mathbb{R}} e^{-4\pi^2 \xi^2 t} e^{2\pi i \xi (x-y)} d\xi \right) dy \\ &= \int_{\mathbb{R}} f(y) \frac{1}{(4\pi t)^{1/2}} \exp\left(-\frac{(x-y)^2}{4t}\right) dy. (*) \end{aligned}$$

Here, $(*)$ is:

$$\begin{aligned} &\int_{\mathbb{R}} e^{-4\pi^2 \xi^2 t} e^{2\pi i x \xi} d\xi \\ &= \int_{\mathbb{R}} e^{2\pi i x \xi - 4\pi^2 \xi^2 t} d\xi \\ &= \int_{\mathbb{R}} e^{-4\pi^2 t \left(\xi^2 - \frac{i(x-y)}{2\pi t} \xi \right)} d\xi \end{aligned}$$

Let

$$u = \frac{i(x-y)}{2\pi t}.$$

Then

$$\int_{\mathbb{R}} e^{-4\pi^2 t (\xi^2 - u \xi)} d\xi$$

Complete the square:

$$\begin{aligned} &= \int_{\mathbb{R}} e^{-4\pi^2 t \left(\left(\xi - \frac{u}{2} \right)^2 - \frac{u^2}{4} \right)} d\xi \\ &= e^{\pi^2 t u^2} \int_{\mathbb{R}} e^{-4\pi^2 t \left(\xi - \frac{u}{2} \right)^2} d\xi \end{aligned}$$

Recall the Gaussian integral:

$$\int_{-\infty}^{\infty} e^{-x^2} dx = \sqrt{\pi},$$

and by translation invariance,

$$\int_{\mathbb{R}} e^{-a(\xi-b)^2} d\xi = \sqrt{\frac{\pi}{a}}.$$

Thus,

$$\begin{aligned} &= e^{\pi^2 t u^2} \sqrt{\frac{\pi}{4\pi^2 t}} \\ &= \frac{1}{(4\pi t)^{1/2}} \exp\left(-\frac{(x-y)^2}{4t}\right) \end{aligned}$$

□

Proposition 6.7. *We claim that*

$$\begin{aligned} &\int_{\mathbb{R}} \frac{1}{(4\pi t)^{1/2}} e^{-\frac{x^2}{4t}} dx = 1 \\ &\lim_{\substack{t \rightarrow 0 \\ |x| \geq \varepsilon > 0}} \frac{1}{(4\pi t)^{1/2}} \int_{\mathbb{R}} e^{-\frac{x^2}{4t}} dx = 0. \end{aligned}$$

Proof. For (1) :

$$\int_{\mathbb{R}} \frac{1}{(4\pi t)^{1/2}} e^{-\frac{x^2}{4t}} dx$$

Let $u = \frac{x}{2\sqrt{t}}$. Then

$$du = \frac{1}{2\sqrt{t}} dx \quad \implies \quad dx = 2\sqrt{t} du.$$

Thus,

$$\begin{aligned} \int_{\mathbb{R}} \frac{1}{(4\pi t)^{1/2}} e^{-\frac{x^2}{4t}} dx &= \int_{\mathbb{R}} \frac{1}{2\sqrt{\pi t}} e^{-u^2} (2\sqrt{t} du) \\ &= \frac{1}{\sqrt{\pi}} \int_{\mathbb{R}} e^{-u^2} du \end{aligned}$$

Using the Gaussian integral

$$\int_{-\infty}^{\infty} e^{-u^2} du = \sqrt{\pi},$$

we obtain

$$\frac{1}{\sqrt{\pi}} \cdot \sqrt{\pi} = 1.$$

For (2) :

We claim that, for $|x| \geq \delta > 0$,

$$e^{-\frac{x^2}{4t}} \lesssim \frac{t}{x^2}.$$

More concretely, we can show that there exists a constant $C > 0$ such that

$$e^{-\frac{x^2}{4t}} \leq C \frac{t}{x^2}, \quad |x| \geq \delta.$$

Indeed, write

$$e^{-\frac{x^2}{4t}} = \frac{t}{x^2} \left(\frac{x^2}{t} e^{-\frac{x^2}{4t}} \right).$$

Let

$$z = \frac{x^2}{t} > 0,$$

so that

$$\frac{x^2}{t} e^{-\frac{x^2}{4t}} = z e^{-z/4}.$$

The function

$$\varphi(z) := z e^{-z/4}, \quad z > 0,$$

is continuous on $(0, \infty)$ and satisfies

$$\lim_{z \rightarrow 0^+} \varphi(z) = 0, \quad \lim_{z \rightarrow \infty} \varphi(z) = 0,$$

so it attains a finite maximum on $(0, \infty)$. Hence

$$C := \sup_{z > 0} z e^{-z/4} < \infty.$$

Therefore, for all $x \in \mathbb{R}$ and $t > 0$,

$$e^{-\frac{x^2}{4t}} = \frac{t}{x^2} (z e^{-z/4}) \leq \frac{t}{x^2} C$$

whenever $x \neq 0$, and in particular for all $|x| \geq \delta > 0$. This proves the desired bound

$$e^{-\frac{x^2}{4t}} \leq C \frac{t}{x^2}, \quad |x| \geq \delta > 0.$$

□

Definition 6.8 [Heat Kernel].

$$H(t, x) = \frac{1}{(4\pi t)^{1/2}} e^{-\frac{x^2}{4t}}$$

is usually called the *heat kernel*.

Proposition 6.9.

$$f \in L^1(\mathbb{R}) \implies \widehat{f} \text{ is continuous}$$

$$\text{So if } \widehat{f} \text{ is continuous and } \widehat{f} \in L^1(\mathbb{R}),$$

then f is equal to a continuous function almost everywhere.

Proof. By proposition 6.6, we know that

$$f(x) := \int_{\mathbb{R}} \widehat{f}(\xi) e^{-4\pi^2 \xi^2 t} e^{2\pi i x \xi} d\xi.$$

We call it $A_t(x)$.

Using the definition of the Fourier transform,

$$\widehat{f}(\xi) = \int_{\mathbb{R}} f(y) e^{-2\pi i y \xi} dy,$$

we substitute this into the integral to obtain

$$A_t(x) = \int_{\mathbb{R}} e^{-4\pi^2 \xi^2 t} e^{2\pi i x \xi} \left(\int_{\mathbb{R}} f(y) e^{-2\pi i y \xi} dy \right) d\xi.$$

By Fubini's theorem (justified since $f \in L^1$ and the Heat kernel is bounded), we may interchange integrals:

$$A_t(x) = \int_{\mathbb{R}} f(y) \left(\int_{\mathbb{R}} e^{-4\pi^2 \xi^2 t} e^{2\pi i (x-y)\xi} d\xi \right) dy.$$

The inner integral is the heat kernel:

$$H_t(x-y) := \int_{\mathbb{R}} e^{-4\pi^2 \xi^2 t} e^{2\pi i (x-y)\xi} d\xi = \frac{1}{\sqrt{4\pi t}} e^{-\frac{(x-y)^2}{4t}}.$$

Thus

$$A_t(x) = \int_{\mathbb{R}} f(y) H_t(x-y) dy.$$

By a change of variables $z = x - y$,

$$\int_{\mathbb{R}} f(y) H_t(x-y) dy = \int_{\mathbb{R}} f(x-z) H_t(z) dz = B_t(x).$$

As $t \rightarrow 0$, we note that

$$A_t(x) = \int_{\mathbb{R}} \widehat{f}(\xi) e^{2\pi i x \xi} e^{-4\pi^2 \xi^2 t} d\xi \longrightarrow \int_{\mathbb{R}} \widehat{f}(\xi) e^{2\pi i x \xi} d\xi.$$

For $B_t(x)$,

$$B_t(x) = \int_{\mathbb{R}} f(x-y) H_t(y) dy,$$

we estimate

$$\int_{\mathbb{R}} \left| \int_{\mathbb{R}} f(x-y) H_t(y) dy - f(x) \right| dx.$$

Using the identity $f(x) = \int_{\mathbb{R}} f(x) H_t(y) dy$, (Indeed, proposition 6.7 gives us the integral of the Heat Kernel is equal to 1, so $f(x) = f(x) * 1 = f(x) * \int_{\mathbb{R}} H_t(y) dy$) we rewrite this as

$$\int_{\mathbb{R}} \left| \int_{\mathbb{R}} (f(x-y) - f(x)) H_t(y) dy \right| dx.$$

Applying Fubini's theorem and the triangle inequality,

$$\leq \int_{\mathbb{R}} \int_{\mathbb{R}} |f(x-y) - f(x)| H_t(y) dy dx. \quad (1)$$

We split the inner integral at a small parameter $\varepsilon > 0$:

$$\int_{\mathbb{R}} |f(x-y) - f(x)| H_t(y) dy = \int_{|y| < \varepsilon} \dots + \int_{|y| \geq \varepsilon} \dots.$$

For $|y| < \varepsilon$,

$|f(x-y) - f(x)|$ is small for a.e. x , (continuity of translations in L^1). (2)

For $|y| \geq \varepsilon$, proposition 6.7 gives us:

$$\int_{|y| \geq \varepsilon} H_t(y) dy \longrightarrow 0 \quad \text{as } t \rightarrow 0. \quad (3)$$

Combining (2) and (3), we conclude that the expression in (1) tends to 0 as $t \rightarrow 0$. □

7 L^p Space

Definition 7.1 [L^p Space]. In general,

$$L_p(A) = \{f \in \mathcal{M}(A) : \|f\|_p < \infty\} / \sim$$

and

$$L_\infty(A) = \{f \in \mathcal{M}(A) : \|f\|_\infty < \infty\} / \sim$$

where $f \sim g$ is the equivalence relation given by $f \sim g \iff f = g$ a.e.

Definition 7.2 [$L^2(\mathbb{R})$]. In this course, we define

$$L^2(\mathbb{R}) = \{f_1 + if_2 | f_i : \mathbb{R} \rightarrow \mathbb{R} \cup \{-\infty, \infty\}, \int |f_i|^2 d\mu < \infty\} / \sim$$

with the inner product:

$$(f, g) = \int_{\mathbb{R}} f(x) \overline{g(x)} = \overline{(g, f)},$$

where $f(x) \overline{g(x)}$ is integrable whenever $f, g \in L^2(\mathbb{R}^d)$ with Cauchy-Schwarz and triangle inequality holds.

Theorem 7.3 (L^2 is complete). $L^2(\mathbb{R}^d)$ is complete in its metric.

Proof. The argument given here follows closely the proof in Chapter 2 that L^1 is complete. Let $\{f_n\}_{n=1}^{\infty}$ be a Cauchy sequence in L^2 , and consider a subsequence $\{f_{n_k}\}_{k=1}^{\infty}$ of $\{f_n\}$ with the following property:

$$\|f_{n_{k+1}} - f_{n_k}\| \leq 2^{-k}, \quad \text{for all } k \geq 1.$$

If we now consider the series whose convergence will be seen below,

$$f(x) = f_{n_1}(x) + \sum_{k=1}^{\infty} (f_{n_{k+1}}(x) - f_{n_k}(x))$$

and

$$g(x) = |f_{n_1}(x)| + \sum_{k=1}^{\infty} |f_{n_{k+1}}(x) - f_{n_k}(x)|,$$

together with the partial sums

$$S_K(f)(x) = f_{n_1}(x) + \sum_{k=1}^K (f_{n_{k+1}}(x) - f_{n_k}(x))$$

and

$$S_K(g)(x) = |f_{n_1}(x)| + \sum_{k=1}^K |f_{n_{k+1}}(x) - f_{n_k}(x)|.$$

then the triangle inequality implies

$$\|S_K(g)\| \leq \|f_{n_1}\| + \sum_{k=1}^K \|f_{n_{k+1}} - f_{n_k}\| \leq \|f_{n_1}\| + \sum_{k=1}^K 2^{-k}.$$

Letting K tend to infinity, and applying the monotone convergence theorem proves that $\int |g|^2 < \infty$, and since $|f| \leq g$, we must have $f \in L^2(\mathbb{R}^d)$.

In particular, the series defining f converges almost everywhere, and since (by construction of the telescopic series) the $(K-1)$ th partial sum of this series is precisely f_{n_K} , we find that

$$f_{n_k}(x) \rightarrow f(x) \quad \text{a.e. } x.$$

To prove that $f_{n_k} \rightarrow f$ in $L^2(\mathbb{R}^d)$ as well, we simply observe that $|f - S_K(f)|^2 \leq (2g)^2$ for all K , and apply the dominated convergence theorem to get $\|f_{n_k} - f\| \rightarrow 0$ as k tends to infinity.

Finally, the last step of the proof consists of recalling that $\{f_n\}$ is Cauchy. Given ε , there exists N such that for all $n, m > N$ we have $\|f_n - f_m\| < \varepsilon/2$. If n_k is chosen so that $n_k > N$, and $\|f_{n_k} - f\| < \varepsilon/2$, then the triangle inequality implies

$$\|f_n - f\| \leq \|f_n - f_{n_k}\| + \|f_{n_k} - f\| < \varepsilon$$

whenever $n > N$. This concludes the proof of the theorem. $\square$

Lemma 7.4 (Parallelogram law). *Parallelogram law:*

$$2\|x\|^2 + 2\|y\|^2 = \|x + y\|^2 + \|x - y\|^2$$

Cauchy-Schwarz inequality:

$$|\langle U, V \rangle| \leq \|U\| * \|V\|$$

In L^2 , it's:

$$|\int_{\mathbb{R}} f(x) \overline{g(x)} dx|^2 \leq \int_{\mathbb{R}} |f(x)|^2 dx \int_{\mathbb{R}} |g(x)|^2 dx$$

Theorem 7.5 (Rietz).

$$\forall \phi \in (L^2(\mathbb{R}))^*, \exists ! g \in L^2(\mathbb{R})$$

such that

$$\phi(f) = \int_{\mathbb{R}} f(x) \overline{g(x)} d\mu, \forall f \in L^2(\mathbb{R})$$

7.1 $(\mathbb{R}^2)^*$

We claim that $(\mathbb{R}^2)^* \cong \mathbb{R}$. One important observation is that $\text{Ker}(\phi)$ is the space of hyper surfaces in $\mathbb{R}^2$ passing $0 \in \mathbb{R}^2$. Given $\phi \in (\mathbb{R}^2)^*$, we look at

$$\omega = v'' + v^\perp$$

where v'' is the projection of ω onto $\text{ker } \phi$, and v is the unit normal vector of $\text{Ker}(\phi)$

To determine ϕ , we only need to know $\phi(v)$. Set $a = \phi(v)$. Then

$$\begin{aligned} \phi(\omega) &= \phi(v'' + v^\perp) \\ &= \phi(v'') \\ &= \phi((\omega, v) v) \\ &= a(\omega, v) \\ &= (\omega, av). \end{aligned}$$

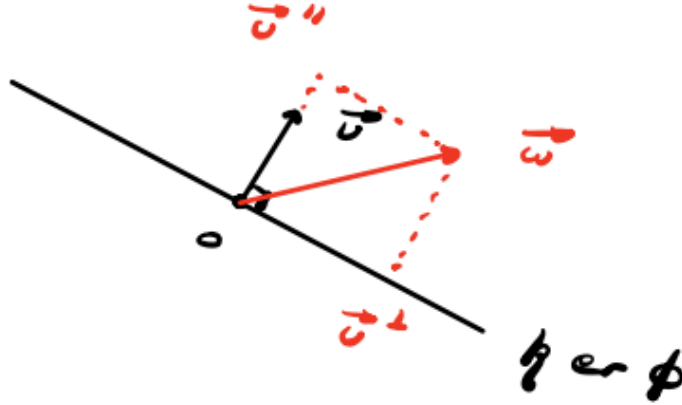


Figure 4: Decomposition of Kernel

where the third step is because v'' is the projection, so $v'' = \text{proj}_v(\omega) = \frac{(v, \omega)}{(v, v)}v$.

Since v is a unit vector, $(v, v) = 1$. So $\phi = av$

Indeed,

$$\forall \omega \in \mathbb{R}^2, \text{ if } \phi(\omega) = 0$$

$$\implies \omega \in \ker \phi \Rightarrow \omega \perp v$$

$$\implies \phi(\omega) = (\omega, \phi(\omega)v) = 0.$$

If $\phi(\omega) \neq 0$, $\exists c \in \mathbb{R}$ such that

$$\phi(\omega - cv) = 0 \tag{*}$$

$$\implies \omega - cv \in \ker \phi$$

$$\implies (\omega, v) = (\omega - cv + cv, v)$$

$$= c$$

$$\implies \phi(\omega) = c\phi(v) = (\omega, v)\phi(v) = (\omega, \phi(v)v).$$

Notice that

$$\forall g_1, g_2 \in (\ker \phi)^\perp, \text{ we know}$$

$$\exists c \text{ such that } \phi(g_1 - cg_2) = 0$$

$$\implies g_1 - cg_2 \in \ker \phi \cap (\ker \phi)^\perp = \{0\}$$

$$\implies \|g_1 - cg_2\| = 0$$

$$\implies g_1 = cg_2.$$

This shows $L^2(\mathbb{R})/\text{Ker}(\phi)$ is 1 dimension.

Theorem 7.6 (Von Neumann). *Let $(\mathbb{R}, \mathcal{M})$ be a measurable space and let μ, ν be σ -finite measures. Then ν may be uniquely decomposed as*

$$d\nu = g d\mu + d\nu_s,$$

where $g \geq 0$ is measurable and $\nu_s \perp \mu$. Moreover $g = \frac{d\nu}{d\mu}$ is the Radon-Nikodym derivative.

Proof. Fix $n \in \mathbb{N}$ and consider the interval $[-n, n]$. Define the Hilbert space

$$H = L^2([-n, n], d\mu + d\nu).$$

This is a Hilbert space with the inner product

$$\langle f, h \rangle_H = \int_{[-n, n]} f(x)h(x) (d\mu + d\nu)(x).$$

Define the linear functional

$$\varphi(f) = \int_{[-n, n]} f d\nu.$$

We check that φ is bounded on H . By Cauchy-Schwarz,

$$|\varphi(f)| = \left| \int f d\nu \right| \leq \left(\int f^2 (d\mu + d\nu) \right)^{1/2} (\nu([-n, n]))^{1/2}.$$

Thus φ is continuous and hence $\varphi \in H^*$.

By the Riesz representation theorem, there exists a unique $g_n \in H$ such that

$$\varphi(f) = \langle f, g_n \rangle_H = \int_{[-n, n]} f(x)g_n(x) (d\mu + d\nu)(x), \quad \forall f \in H.$$

In particular,

$$\int_{[-n, n]} f d\nu = \int_{[-n, n]} f(x)g_n(x) (d\mu + d\nu)(x). \quad (*)$$

Claim: $0 \leq g_n \leq 1$ $(\mu + \nu)$ -a.e. Indeed, take $f = \chi_E$ for a measurable $E \subseteq [-n, n]$ with $(\mu + \nu)(E) > 0$. Then $(*)$ gives

$$\nu(E) = \int_E g_n (d\mu + d\nu).$$

Divide both sides by $(\mu + \nu)(E)$ to obtain

$$0 \leq \frac{\nu(E)}{(\mu + \nu)(E)} = \frac{\int_E g_n (d\mu + d\nu)}{(\mu + \nu)(E)} \leq 1.$$

The left-hand side equals the average value of g_n on E . Since this holds for all measurable E , we conclude

$$0 \leq g_n \leq 1 \quad (\mu + \nu)\text{-a.e.}$$

and therefore also μ -a.e.

Define the sets

$$R = \{x \in [-n, n] : 0 \leq g_n(x) < 1\}, \quad S = \{x \in [-n, n] : g_n(x) = 1\}.$$

Define measures

$$\nu_s(E) := \nu(E \cap S), \quad \nu_a(E) := \nu(E \cap R).$$

Step 1: $\nu_s \perp \mu$.

Let $E \subseteq S$. From $(*)$ with $f = \chi_E$,

$$\nu(E) = \int_E g_n (d\mu + d\nu) = \int_E 1 \cdot (d\mu + d\nu) = \mu(E) + \nu(E).$$

Subtract $\nu(E)$ to obtain $\mu(E) = 0$. Thus every subset of S has μ -measure 0 and hence $\nu_s \perp \mu$.

Step 2: Identify the absolutely continuous part.

For $E \subseteq R$, again using $(*)$:

$$\nu(E) = \int_E g_n (d\mu + d\nu) = \int_E g_n d\mu + \int_E g_n d\nu.$$

Rearrange:

$$\int_E (1 - g_n) d\nu = \int_E g_n d\mu. \quad (**)$$

We want to express $\nu(E)$ in terms of μ . Formally one might hope

$$\frac{g_n}{1 - g_n} \quad \text{acts like} \quad \frac{d\nu}{d\mu}.$$

But $\frac{1}{1-g_n}$ might not be in L^2 . So we approximate.
Define

$$h_k = \sum_{j=0}^k g_n^j.$$

Then $h_k \uparrow \frac{1}{1-g_n}$ pointwise on R . Put $f = \chi_E h_k \in L^2([-n, n], d\mu + d\nu)$. Applying (**) repeatedly,

$$\int_E (1 - g_n) h_k d\nu = \sum_{j=0}^k \int_E (1 - g_n) g_n^j d\nu = \sum_{j=0}^k \int_E g_n^{j+1} d\mu = \int_E \left(\sum_{j=0}^k g_n^{j+1} \right) d\mu.$$

Take $k \rightarrow \infty$ and apply Monotone Convergence to get

$$\int_E (1 - g_n) \left(\frac{1}{1 - g_n} \right) d\nu = \int_E \frac{g_n}{1 - g_n} d\mu.$$

Since $(1 - g_n) \frac{1}{1 - g_n} = 1$ on R ,

$$\nu(E) = \int_E \frac{g_n}{1 - g_n} d\mu.$$

Thus on R ,

$$\frac{d\nu}{d\mu} = \frac{g_n}{1 - g_n}.$$

Let $g = \frac{g_n}{1 - g_n}$ on R and $g = 0$ on S . Then

$$\nu_a(E) = \int_E g d\mu.$$

Putting everything together,

$$\nu(E) = \nu_a(E) + \nu_s(E) = \int_E g d\mu + \nu(E \cap S),$$

with $\nu_s \perp \mu$.

Finally take $n \rightarrow \infty$ and use monotone convergence to get the full decomposition on $\mathbb{R}$. □

Definition 7.7 [Schwartz Class]. A function $f \in S$, where S is the Schwartz class if

$$|x^\alpha \frac{d}{dx^\beta} f| \leq c < \infty$$

We claim that $f \in S \implies \hat{f} \in S$

Proof. Let's check $\frac{d}{d\xi^\alpha} \hat{f}$.

$$\begin{aligned}
\frac{d}{d\xi^\alpha} \hat{f} &= \frac{d}{d\xi^\alpha} \int_{\mathbb{R}} f(x) e^{2\pi i x \xi} dx \\
&= \int_{\mathbb{R}} (2\pi i x)^\alpha f(x) e^{2\pi i x \xi} dx \\
(2\pi i \xi)^\beta \frac{d}{d\xi^\alpha} \hat{f} &= \int_{\mathbb{R}} (2\pi i x)^\alpha f(x) (2\pi i \xi)^\beta e^{2\pi i x \xi} dx \\
&= \int_{\mathbb{R}} (2\pi i x)^\alpha f(x) \frac{d^\beta}{dx^\beta} e^{2\pi i x \xi} dx \\
&= \int_{\mathbb{R}} (-1)^\beta \frac{d^\beta}{dx^\beta} ((2\pi i x)^\alpha f(x)) e^{2\pi i x \xi} dx \\
&\implies \left| \xi^\beta \frac{d}{d\xi^\alpha} \hat{f} \right| \leq C < \infty
\end{aligned}$$

□

Lemma 7.8 (Plancherel Identity). *If $f \in S$, then*

$$\|f\|_{L^2(\mathbb{R})} = \|\hat{f}\|_{L^2(\mathbb{R})},$$

where S is the Schwartz class.

Proof.

$$\begin{aligned}
\|\hat{f}\|_{L^2(\mathbb{R})}^2 &= \int_{\mathbb{R}} \hat{f} \overline{\hat{f}} d\mu = \int_{\mathbb{R}} f \overline{\hat{\hat{f}}} d\mu = \int_{\mathbb{R}} \hat{f} \overline{f} d\mu = \int_{\mathbb{R}} f \overline{f} d\mu = \|f\|_{L^2(\mathbb{R})}^2. \\
&\implies \|\hat{f}\|_{L^2(\mathbb{R})} = \|f\|_{L^2(\mathbb{R})}.
\end{aligned}$$

□

Example 7.9 [Bump Function]. Let

$$f_N = f \chi_{\{|x| < N, |f| < N\}}$$

We claim $f_N \rightarrow f$ in $L^2(\mathbb{R})$

Proof. It suffices to approximate f_N . Consider a smooth mollifier ρ with the properties

$$\rho \in C_c^\infty(\mathbb{R}), \quad \rho \geq 0, \quad \int_{\mathbb{R}} \rho(z) dz = 1.$$

For example,

$$\rho(x) = \begin{cases} C e^{-\frac{1}{1-x^2}}, & |x| < 1, \\ 0, & |x| \geq 1, \end{cases} \quad C^{-1} = \int_{-1}^1 e^{-\frac{1}{1-z^2}} dz.$$

Define the ε -scaled mollifier

$$\rho_\varepsilon(x) = \frac{1}{\varepsilon} \rho\left(\frac{x}{\varepsilon}\right).$$

Consider ρ_ε smooth

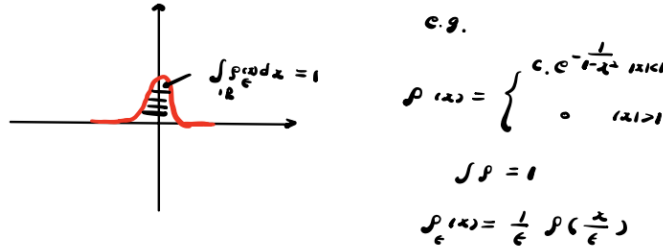


Figure 5: Bump function

Definition. Define the mollification of f_N by

$$f_N^\varepsilon(x) = \rho_\varepsilon\text{-weighted average of } f_N = \int_{\mathbb{R}} f_N(y) \rho_\varepsilon(x - y) dy.$$

Properties.

- f_N^ε is smooth with compact support, since ρ_ε is smooth and compactly supported.
- f_N^ε is uniformly bounded in ε :

$$|f_N^\varepsilon(x)| \leq \int_{\mathbb{R}} |f_N(y)| \rho_\varepsilon(x - y) dy \leq \|f_N\|_{L^\infty} \leq N.$$

- Derivatives satisfy

$$\partial_x f_N^\varepsilon(x) = \int_{\mathbb{R}} f_N(y) \partial_x (\rho_\varepsilon(x - y)) dy.$$

Since $\rho'_\varepsilon(z) = \frac{1}{\varepsilon^2} \rho'\left(\frac{z}{\varepsilon}\right)$ and ρ' is bounded with compact support, all derivatives $\partial_x^k f_N^\varepsilon$ exist and are bounded.

□

8 Hardy Space

Definition 8.1 [Hardy Space]. Let $\mathbb{R}_+^2 = \{z = x + iy, x \in \mathbb{R}, y > 0\}$ be the upper half-plane. We define the *Hardy space* $H^2(\mathbb{R}_+^2)$ to consist of all functions F analytic in $\mathbb{R}_+^2$ with the property that

$$\sup_{y>0} \int_{\mathbb{R}} |F(x + iy)|^2 dx < \infty. \quad (2)$$

Example 8.2 [Hardy Space]. Let us first describe a (typical) example of a function $F \in H^2(\mathbb{R}_+^2)$. We start with a function $\widehat{F}_0$ that belongs to $L^2(0, \infty)$, and write

$$F(x + iy) = \int_0^\infty \widehat{F}_0(\xi) e^{2\pi i \xi z} d\xi, \quad z = x + iy, y > 0. \quad (6)$$

(The choice of the particular notation $\widehat{F}_0$ will become clearer below.) We claim that for any $\delta > 0$ the integral (6) converges absolutely and uniformly as long as $y \geq \delta$. Indeed,

$$|\widehat{F}_0(\xi) e^{2\pi i \xi z}| = |\widehat{F}_0(\xi)| e^{-2\pi \xi y},$$

hence by the Cauchy–Schwarz inequality,

$$\int_0^\infty |\widehat{F}_0(\xi) e^{2\pi i \xi z}| d\xi \leq \left(\int_0^\infty |\widehat{F}_0(\xi)|^2 d\xi \right)^{1/2} \left(\int_0^\infty e^{-4\pi \xi \delta} d\xi \right)^{1/2},$$

from which the asserted convergence is established. From the uniform convergence it follows that $F(z)$ is holomorphic in the upper half-plane.

Moreover, by Plancherel’s theorem,

$$\int_{\mathbb{R}} |F(x + iy)|^2 dx = \int_0^\infty |\widehat{F}_0(\xi)|^2 e^{-4\pi \xi y} d\xi \leq \|\widehat{F}_0\|_{L^2(0, \infty)}^2.$$

and in fact, by the monotone convergence theorem,

$$\sup_{y>0} \int_{\mathbb{R}} |F(x + iy)|^2 dx = \|\widehat{F}_0\|_{L^2(0, \infty)}^2.$$

In particular, F belongs to $H^2(\mathbb{R}_+^2)$. The main result we prove next is the converse, that is, every element of the space $H^2(\mathbb{R}_+^2)$ is in fact of the form (6).

8.1 Fourier Transform

For a function

$$f_y(x) = \frac{1}{x + i(y + 1)},$$

consider its fourier transform:

$$\hat{f}_y(\xi) = \int_{\mathbb{R}} \frac{1}{x + i(y+1)} e^{-2\pi i x \xi} dx$$

We claim that

$$\hat{f}_y(\xi) = \begin{cases} -2\pi i e^{-2\pi \xi(y+1)}, & \xi > 0, \\ 0, & \xi < 0, \\ -\pi i, & \xi = 0. \end{cases}$$

Proof. The key is to understand

$$\frac{1}{z + i}$$

is the function

$$-2\pi i e^{-2\pi \xi} \chi_{\{\xi > 0\}} = \hat{f}_0(\xi)$$

when $\xi > 0$

What is this function? It is the Fourier transform

$$\hat{f}_0(\xi) = \int_{\mathbb{R}} \frac{1}{x + i} e^{2\pi i x \xi} dx.$$

Also,

$$\|\hat{f}_0\|_{L^2(0, \infty)}^2 = \int_0^\infty (4\pi^2) e^{-4\pi \xi} d\xi = \pi.$$

□

8.1.1 Computation of Fourier Transform

- Let's compute this for $\xi < 0$. From the graph below, we know that the path can be treated as $[-R, R] + C_R$, so we can split the integral into two parts. Together, they formed a closed loop.

For a closed loop γ ,

$$\gamma : [0, 1] \rightarrow \mathbb{C}, \quad \gamma(0) = \gamma(1).$$

We compute

$$\oint_{\gamma} z^n dz = \int_0^1 (\gamma(t))^n \gamma'(t) dt.$$

Let

$$F(z) = \frac{z^{n+1}}{n+1}, \quad \text{so that} \quad F'(z) = z^n.$$

Then by the (real) Fundamental Theorem of Calculus applied to $F(\gamma(t))$,

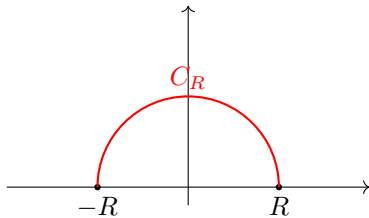
$$\int_0^1 (\gamma(t))^n \gamma'(t) dt = \int_0^1 F'(\gamma(t)) \gamma'(t) dt = F(\gamma(1)) - F(\gamma(0)).$$

Since γ is a closed loop, $\gamma(1) = \gamma(0)$, hence

$$F(\gamma(1)) - F(\gamma(0)) = 0,$$

and therefore

$$\oint_{\gamma} z^n dz = 0.$$



Key: Deformation invariance

$$\int_{-R}^R + \int_{C_R} = 0$$

$$|e^{-2\pi i z \xi}| = |e^{-2\pi i (x+iy)\xi}| = e^{-2\pi y \xi}$$

Since $\xi < 0$, we have $-2\pi y \xi > 0$ when $y > 0$, and therefore

$$e^{-2\pi y \xi} \text{ decays fast as } y \rightarrow +\infty.$$

Thus

$$\int_{C_R} \longrightarrow 0 \quad \text{as } R \rightarrow \infty.$$

Therefore,

$$\int_{-R}^R \longrightarrow 0 \quad \text{as } R \rightarrow \infty.$$

- $\xi = 0$:

We study

$$\lim_{R \rightarrow \infty} \int_{-R}^R \frac{1}{x + i(1 + y_0)} dx.$$

Since

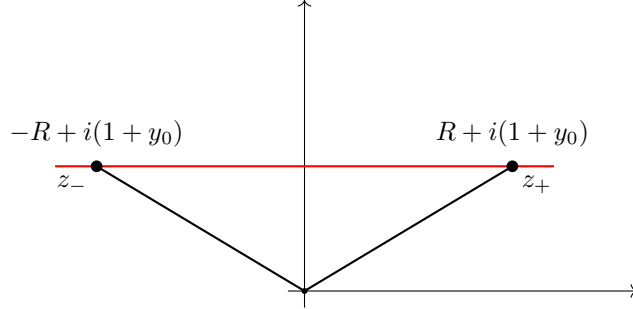
$$\int \frac{dx}{x + i(1 + y_0)} = \ln(x + i(1 + y_0)),$$

we obtain

$$\lim_{R \rightarrow \infty} \int_{-R}^R \frac{1}{x + i(1 + y_0)} dx = \lim_{R \rightarrow \infty} [\ln(x + i(1 + y_0))]_{x=-R}^{x=R} = \lim_{R \rightarrow \infty} (\ln(z_+) - \ln(z_-)),$$

where

$$z_{\pm} = \pm R + i(1 + y_0).$$



We write each endpoint in polar form: (assume symmetric. That is, $z_+ = z_-$. Then the difference will be the difference of the angles. One has angle 0 degree while the other has π)

$$z_{\pm} = r e^{i\theta_{\pm}} \Rightarrow \ln z_{\pm} = \ln r + i\theta_{\pm}.$$

As $R \rightarrow \infty$, the modulus is the same:

$$|z_+| = |z_-| = r,$$

but the arguments satisfy

$$\theta_+ - \theta_- \longrightarrow -\pi.$$

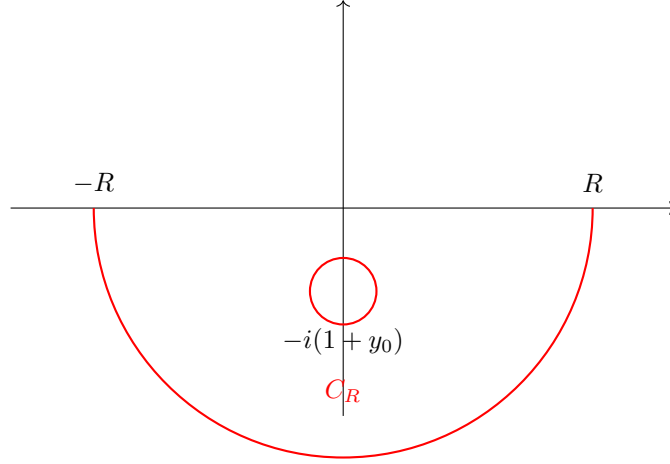
Therefore

$$\ln z_+ - \ln z_- = (\ln r + i\theta_+) - (\ln r + i\theta_-) = i(\theta_+ - \theta_-) \longrightarrow -\pi i.$$

$$\boxed{\lim_{R \rightarrow \infty} \int_{-R}^R \frac{dx}{x + i(1 + y_0)} = -\pi i.}$$

- $\xi > 0$

$$\left| \frac{1}{z + i(1 + y_0)} e^{-2\pi i z \xi} \right| = e^{-2\pi y \xi}, \quad (y < 0 \text{ on the lower semicircle}).$$



Since the integrand is analytic away from $z = -i(1 + y_0)$, the contour decomposes:

$$\int_{(-R, R)} + \int_{C_R} = \int_{C_\epsilon},$$

where C_ϵ is a small circle around the pole.

As $R \rightarrow \infty$, the lower semicircle satisfies

$$\int_{C_R} \rightarrow 0 \quad (e^{-2\pi y \xi} \text{ decays exponentially}).$$

Hence

$$\int_{-\infty}^{\infty} \frac{e^{-2\pi i x \xi}}{x + i(1 + y_0)} dx = \int_{C_\epsilon} \frac{e^{-2\pi i z \xi}}{z + i(1 + y_0)} dz.$$

We rewrite the integrand around the pole:

$$\int_{C_\epsilon} \frac{e^{-2\pi i z \xi}}{z - (-i(1 + y_0))} dz = \int_{C_\epsilon} \frac{e^{-2\pi i z \xi} - e^{-2\pi i (-i(1 + y_0)) \xi}}{z - (-i(1 + y_0))} dz + \int_{C_\epsilon} \frac{e^{-2\pi i (-i(1 + y_0)) \xi}}{z - (-i(1 + y_0))} dz.$$

Call these two integrals A and B :

$$A = \int_{C_\epsilon} \frac{e^{-2\pi i z \xi} - e^{-2\pi i (-i(1 + y_0)) \xi}}{z - (-i(1 + y_0))} dz, \quad B = \int_{C_\epsilon} \frac{e^{-2\pi i (-i(1 + y_0)) \xi}}{z - (-i(1 + y_0))} dz.$$

Evaluation of A

The numerator vanishes at the pole, so shrinking $C_\epsilon \rightarrow -i(1+y_0)$:

$$A = 0.$$

Evaluation of B

Using the Fundamental Theorem of Calculus for contour integrals:

$$B = e^{-2\pi i(-i(1+y_0))\xi} \int_{C_\epsilon} \frac{dz}{z - (-i(1+y_0))} = e^{-2\pi i(-i(1+y_0))\xi} (-2\pi i).$$

Since

$$-2\pi i(-i(1+y_0))\xi = -2\pi(1+y_0)\xi,$$

we obtain

$$B = -2\pi i e^{-2\pi(1+y_0)\xi}.$$

$$\boxed{\int_{-\infty}^{\infty} \frac{e^{-2\pi i x \xi}}{x + i(1+y_0)} dx = -2\pi i e^{-2\pi(1+y_0)\xi}, \quad \xi > 0.}$$

8.2 $H^2(\mathbb{R}_+^2) \cong L^2((0, \infty))$

Let us build

$$H^2(\mathbb{R}^+) \cong L^2((0, \infty)) \text{ abstractly.}$$

For this, we start with

$$\widehat{f}_0 \in L^2((0, \infty)).$$

Set

$$f(z) = \int_0^\infty \widehat{f}_0(\xi) e^{2\pi i z \xi} d\xi.$$

Goal:

- (1) $f(z) \in H^2(\mathbb{R}^+)$,
- (2) Any $f \in H^2(\mathbb{R}^+)$ arises in this way.

For (1):

$$f'(z) = \int_0^\infty \widehat{f}_0(\xi) (2\pi i \xi) e^{2\pi i z \xi} d\xi$$

Let $f_y(x) := f(x + iy)$. Then

$$\int_{\mathbb{R}} |f_y(x)|^2 dx = \int_0^\infty |\widehat{f}_0(\xi)|^2 e^{-4\pi y \xi} d\xi \quad (\text{Plancherel}).$$

Hence

$$\sup_{y>0} \int_{\mathbb{R}} |f_y(x)|^2 dx = \int_0^\infty |\widehat{f}_0(\xi)|^2 d\xi.$$

9 $L^p(\infty)$ Space

Here we use the metric:

$$\|f - g\|_p = \left(\int |f - g|^p dx \right)^{1/p}$$

Recall the definition of a metric:

Definition 9.1 [$\|\cdot\|_p$ Norm]. $\|\cdot\|_p$ is a norm if it satisfies:

1. $\|\alpha f\|_p = |\alpha| \|f\|_p$
2. $\|f\|_p = 0 \iff f = 0$ a.e.
3. $\|f + g\|_p \leq \|f\|_p + \|g\|_p$ (Minkowski)

Lemma 9.2 (Hölder's Inequality). Let $(X, \mathcal{B}, \mu)$ be a measure space. Let $1 < p < \infty$ and define q so that

$$\frac{1}{p} + \frac{1}{q} = 1.$$

If $f \in L^p(\mu)$ and $g \in L^q(\mu)$, then $fg \in L^1(\mu)$ and

$$\|fg\|_1 \leq \|f\|_p \|g\|_q.$$

Equality holds if and only if $|f|^p$ and $|g|^q$ are collinear.

Proof. WLOG assume $\|f\|_p = 1$ and $\|g\|_q = 1$. We need to show

$$\int_{\mathbb{R}} |fg| \leq 1.$$

This follows from

$$|fg| \leq \frac{|f|^p}{p} + \frac{|g|^q}{q} \quad (*)$$

so that

$$\int_{\mathbb{R}} |fg| \leq \int_{\mathbb{R}} \left(\frac{|f|^p}{p} + \frac{|g|^q}{q} \right) = \frac{1}{p} + \frac{1}{q} = 1.$$

For $(*)$, it suffices to show

$$A^\theta B^{1-\theta} \leq \theta A + (1-\theta)B, \quad 0 \leq \theta \leq 1, A \geq 0, B \geq 0.$$

WLOG: set $B = 1, 0 \leq A \leq 1$. We need to show

$$A^\theta \leq \theta A + (1-\theta), \quad 0 \leq A \leq 1.$$

Let

$$h(A) = A^\theta - \theta A - (1-\theta).$$

Then

$$h'(A) = \theta A^{\theta-1} - \theta = \theta(A^{\theta-1} - 1) \geq 0$$

since $0 \leq A \leq 1$ and $\theta - 1 \leq 0$, so $A^{\theta-1} \geq 1$.

Thus h is increasing, hence

$$h(A) \leq h(1) = 0.$$

Therefore

$$A^\theta \leq \theta A + (1 - \theta), \quad 0 \leq A \leq 1,$$

as required. □

Definition 9.3 [Legendre transform]. Let $I \subset \mathbb{R}$ be an interval, and $f : I \rightarrow \mathbb{R}$ a **convex** function; then the Legendre transform of f is the function $f^* : I^* \rightarrow \mathbb{R}$ defined by

$$f^*(x^*) = \sup_{x \in I} (x^*x - f(x)), \quad I^* = \left\{ x^* \in \mathbb{R} : \sup_{x \in I} (x^*x - f(x)) < \infty \right\},$$

where $\sup$ denotes the supremum over I , e.g., $x \in I$ is chosen such that $x^*x - f(x)$ is maximized at each x^* , or x^* is such that $x^*x - f(x)$ has a bounded value throughout I (e.g., when $f(x)$ is a linear function).

Definition 9.4 [Dual Space]. Let $1 \leq p < \infty$ and let q be the conjugate exponent, $1/p + 1/q = 1$. The dual space of $L^p(\mathbb{R})$, denoted $(L^p(\mathbb{R}))^*$, is the set of all **bounded linear functionals** on $L^p(\mathbb{R})$. By the Riesz–Markov–Kakutani representation, every $\Phi \in (L^p(\mathbb{R}))^*$ has the form

$$\Phi(f) = \int_{\mathbb{R}} f(x)g(x) dx,$$

for a unique $g \in L^q(\mathbb{R})$. Moreover,

$$\|\Phi\| = \|g\|_{L^q},$$

Theorem 9.5 (TFAE). *The following statements are equivalent:*

1. ϕ is continuous.
2. ϕ is bounded.
3. $\ker \phi$ is closed.

Proof. 1. (1) $\Rightarrow$ (3) Otherwise, $\phi(f_i) \rightarrow \infty$ with $\|f_i\|_p = 1$.

$$\frac{f_i}{\phi(f_i)} \rightarrow 0 \quad \text{but} \quad \phi\left(\frac{f_i}{\phi(f_i)}\right) = 0.$$

2. (3) $\Rightarrow$ (1)

$$|\phi(f_i - f)| \leq C\|f_i - f\|_p.$$

3. (2) $\Rightarrow$ (3) Suppose $\phi(f_i) = 0$ and $f_i \rightarrow f$. Then

$$|\phi(f)| = |\phi(f - f_i)| \leq C\|f - f_i\|_p$$

so

$$\phi(f) = 0.$$

4. (3) $\Rightarrow$ (2) WLOG assume $\phi \neq 0$. Pick f so that $\phi(f) = 1$. If ϕ is not bounded, then $\phi(f_i) \rightarrow \infty$ with $\|f_i\|_p = 1$.

Consider

$$g = f - \frac{f_i}{\phi(f_i)}.$$

Then

$$\phi(g) = 0$$

but

$$g = f - \frac{f_i}{\phi(f_i)} \rightarrow f,$$

so

$$\phi(f) = 0,$$

a contradiction. $\circlearrowleft$

□

Definition 9.6 [Function in $(L^p(\mathbb{R}))^*$]. $\phi \in (L^p(\mathbb{R}))^*$ means

- ϕ is linear;
- ϕ satisfies theorem 9.5

Definition 9.7 [Absolutely continuous]. Suppose f is integrable on $\mathbb{R}^d$. Then for every $\epsilon > 0$, there is $\delta > 0$ with $\mu(E) < \delta$ st

$$\int_E |f| < \epsilon$$

Theorem 9.8 (Interchanging sum and integral: Tonelli and Fubini). *Let $(X, \mathcal{A}, \mu)$ be a measure space, and let $(f_k)_{k \in \mathbb{N}}$ be a sequence of $\mathcal{A}$ -measurable functions $f_k : X \rightarrow [-\infty, \infty]$.*

Tonelli: If $f_k(x) \geq 0$ for all k and all $x \in X$, then

$$\int_X \sum_{k=1}^{\infty} f_k(x) d\mu(x) = \sum_{k=1}^{\infty} \int_X f_k(x) d\mu(x) \in [0, \infty].$$

Fubini (for series): If

$$\sum_{k=1}^{\infty} \int_X |f_k(x)| d\mu(x) < \infty,$$

then each f_k is integrable, the series $\sum_{k=1}^{\infty} f_k(x)$ converges absolutely (i.e. $\sum_{k=1}^{\infty} \int_X |f_k(x)| d\mu(x) < \infty$,) for μ -a.e. $x \in X$, the sum

$$f(x) := \sum_{k=1}^{\infty} f_k(x)$$

defines an integrable function f , and

$$\int_X f(x) d\mu(x) = \int_X \sum_{k=1}^{\infty} f_k(x) d\mu(x) = \sum_{k=1}^{\infty} \int_X f_k(x) d\mu(x).$$

Theorem 9.9 $((L^p)^* \cong L^q)$.

$$(L^p(\mathbb{R}))^* \cong L^q(\mathbb{R}), \quad 1 < p < \infty.$$

i.e., for every $\phi : L^p(\mathbb{R}) \rightarrow \mathbb{R}$ with

1. $\phi(af + bg) = a\phi(f) + b\phi(g)$
2. $|\phi(f)| \leq C\|f\|_p$

Proof. Here our goal is to find $g \in L^q(\mathbb{R})$ so that

$$\Phi(f) = \int_{\mathbb{R}} fg d\mu,$$

Note the norm defined on the dual space $L^p(\mathbb{R})^*$ is:

$$\|\Phi\| = \sup_{\|f\|_p=1} |\phi(f)|$$

We claim that $\|\phi\| = \|g\|_q$, here's the proof:

Since

$$|\Phi(f)| = \left| \int_{\mathbb{R}} f(x)g(x) dx \right| \leq \|f\|_p \|g\|_q \quad (\text{H\"older}),$$

By assuming $\|f\|_p = 1$ (otherwise take $f' = \frac{f}{\|f\|_p}$), we have

$$\|\Phi\| \leq \|g\|_q.$$

To prove the reverse inequality, choose

$$f = c \operatorname{sgn}(g) |g|^{q-1},$$

where the constant $c > 0$ is chosen so that $\|f\|_p = 1$.

Compute $\|f\|_p$:

$$\|f\|_p^p = \int_{\mathbb{R}} |c|^p |g|^{(q-1)p} dx.$$

Using the identity $(q-1)p = q$, we get

$$\|f\|_p^p = c^p \int_{\mathbb{R}} |g|^q dx = c^p \|g\|_q^q.$$

To make $\|f\|_p = 1$, choose

$$c = \|g\|_q^{1-q/p} = \|g\|_q^{1-q/p} = \|g\|_q^{1-q/p}.$$

But since $1/p + 1/q = 1$, we have $q/p = q-1$, hence

$$c = \|g\|_q^{1-q} = \|g\|_q^{-(q-1)}.$$

Compute $\Phi(f)$:

$$\Phi(f) = \int_{\mathbb{R}} fg = c \int_{\mathbb{R}} |g|^q dx = c \|g\|_q^q = \|g\|_q.$$

Thus

$$\|\Phi\| \geq |\Phi(f)| = \|g\|_q,$$

since $\|\Phi\|$ is the sup by definition.

Combining both inequalities gives

$$\boxed{\|\Phi\| = \|g\|_q}.$$

Now we find such g : Set $v(E) = \phi(\chi_E)$, where $E \subset [-n, n]$. We claim that v is a signed measure, here's the proof:

Indeed, $v(\emptyset) = \phi(\chi_{\emptyset}) = \phi(0) = 0$. Notice that

$$v\left(\sum_{i=1}^n E_i\right) = \phi(\chi_{\sum_{i=1}^n E_i}) = \sum_{i=1}^n v(E_i)$$

We then need to show

$$\lim_{n \rightarrow \infty} \nu\left(\bigcup_{k=1}^n E_k\right) = \nu\left(\bigcup_{k=1}^{\infty} E_k\right).$$

$$\text{Set } X_N = \sum_{k=1}^N \chi_{E_k}, \quad X_{\infty} = \sum_{k=1}^{\infty} \chi_{E_k}.$$

Note since χ_{E_k} is either 1 or 0, in both cases, $\chi_{E_k}^p = \chi_{E_k}$.

Then

$$\begin{aligned}
\|X_N - X_\infty\|_p &= \left(\int \sum_{k=N+1}^{\infty} \chi_{E_k}^p d\mu \right)^{1/p} \\
&= \left(\int \sum_{k=N+1}^{\infty} \chi_{E_k} d\mu \right)^{1/p} \\
&= \left(\sum_{k=N+1}^{\infty} \mu(E_k) \right)^{1/p} \quad (*) \\
&= \left(\mu \left(\bigcup_{k=N+1}^{\infty} E_k \right) \right)^{1/p} \xrightarrow{N \rightarrow \infty} 0.
\end{aligned}$$

Here $(*)$ is valid since $\chi_{E_k} \geq 0$ so we can apply the Fubini's theorem (or DCT) to interchange the integral and summation. By definition of Lebesgue measure on a measurable set, $\int \chi_{E_k} d\mu = \mu(E_k)$.

Note that $\nu \ll \mu$.

By the Radon–Nikodym theorem,

$$\nu(E) = \varphi(\chi_E) = \int_E g d\mu = \int g_n \chi_E d\mu,$$

where $g = \frac{d\nu}{d\mu}$ and $\int |g_n| < \infty$.

Let $n \rightarrow \infty$, this implies

$$\phi(f) = \int g f d\mu,$$

where $f \in L^p(\mathbb{R})$.

Now the question becomes why $\|g\|_{L^q(\mathbb{R})} < \infty$?

Take s_k simple with $|s_k| \leq |g|$ and $s_k \rightarrow g$.

Let

$$t_k = c |s_k|^{q-1} \operatorname{sgn}(g), \quad \int |t_k|^p d\mu = 1.$$

Then

$$|\varphi(t_k)| \leq c.$$

But

$$\varphi(t_k) = \int g t_k d\mu = \int |s_k|^{q-1} |g| d\mu.$$

By Fatou's lemma,

$$\int |g|^p d\mu \leq \liminf_{k \rightarrow \infty} \int |s_k|^{p-1} |g| d\mu \leq c.$$

Thus,

$$(L^p(\mathbb{R}))^* \cong L^q(\mathbb{R}), \quad 1 < p < \infty.$$

□

It is natural to think about the dual space of the dual space? Is it isomorphic to the original space?

Theorem 9.10 (Riesz Representation Theorem, I). *Let $\Gamma \in L^p(X, \mu)^*$, where $1 \leq p < \infty$ and μ is σ -finite. Then if $\frac{1}{p} + \frac{1}{q} = 1$, there exists a unique $g \in L^q(X, \mu)$ such that*

$$\Gamma(f) = \int_X fg \, d\mu = \phi_g(f)$$

for all $f \in L^p(X, \mu)$. Moreover,

$$\|\Gamma\| = \|g\|_q.$$

Proof. Assume that μ is finite. Then every bounded measurable function is in $L^p(X, \mu)$. Define a $\lambda : \mathcal{A} \rightarrow \mathbb{R}$ by

$$\lambda(E) = \Gamma(\chi_E).$$

Let $\{E_n\} \subseteq \mathcal{A}$ be a sequence of disjoint sets, and let $E = \bigcup_{n=1}^{\infty} E_n$. Let $\alpha_n = \text{sgn}(\chi_{E_n})$ and $f = \sum_{n=1}^{\infty} \alpha_n \chi_{E_n}$. Then $f \in L^p(X, \mu)$ and

$$\Gamma(f) = \sum_{n=1}^{\infty} \lambda(E_n) < \infty \quad \text{and so} \quad \sum_{n=1}^{\infty} |\lambda(E_n)| = \Gamma(\chi_E) = \lambda(E).$$

Therefore λ is a finite signed measure. Clearly, if $\mu(E) = 0$ then $\chi_E = 0$ almost everywhere, so $\lambda(E) = 0$. Therefore $\lambda \ll \mu$. By the Radon–Nikodym theorem, there is an integrable function g such that

$$\lambda(E) = \int_E g \, d\mu \quad \text{for all } E \in \mathcal{A}.$$

If φ is simple, then $\Gamma(\varphi) = \int \varphi g \, d\mu$ by linearity of the integral. But

$$|\Gamma(\varphi)| \leq \|\Gamma\| \|\varphi\|_p \quad \text{for all simple functions } \varphi,$$

so $g \in L^q(X, \mu)$ by the lemma above. Now $\Gamma - \phi_g \in L^p(X, \mu)^*$ and $\Gamma - \phi_g = 0$ on the space of simple functions. Since simple functions are dense in $L^p(X, \mu)$,

$$\Gamma = \phi_g.$$

We have that $\|\Gamma\| = \|g\|_q$ as before.

Now assume that μ is σ -finite. We can write $X = \bigcup_{n=1}^{\infty} X_n$, where $\mu(X_n) < \infty$ and $X_n \subseteq X_{n+1}$ for all $n \geq 1$. For each $n \geq 1$, the proof above gives $g_n \in L^q(X, \mu)$, vanishing outside X_n , such that

$$\Gamma(f) = \int fg \, d\mu \quad \text{for all } f \in L^p(X, \mu) \text{ vanishing outside } X_n.$$

Moreover, $\|g_n\|_q \leq \|\Gamma\|$. By the uniqueness of the g_n 's, we can assume that $g_{n+1} = g_n$ on X_n . Let $g(x) = \lim_{n \rightarrow \infty} g_n(x)$. We have that $|g_n| \nearrow |g|$. By the Monotone Convergence Theorem,

$$\int |g|^q d\mu = \lim_{n \rightarrow \infty} \int |g_n|^q d\mu \leq \|\Gamma\|_q.$$

Hence $g \in L^q(X, \mu)$. Let $f \in L^p(X, \mu)$ and let $f_n = f\chi_{X_n}$. Then $f_n \rightarrow f$ pointwise and $f_n \in L^p(X, \mu)$ for all $n \geq 1$. Since $|fg| \in L^1(X, \mu)$ and $f_n g \leq |fg|$, the Lebesgue Dominated Convergence Theorem shows

$$\int fg d\mu = \lim_{n \rightarrow \infty} \int f_n g d\mu = \lim_{n \rightarrow \infty} \Gamma(f_n) = \Gamma(f).$$

If $p = 1$, then we cannot drop the assumption of σ -finiteness. $\square$

Theorem 9.11 $((L^p(\mu))^{**} \cong L^p(\mu) \iff 1 < p < \infty)$. Let $(\Omega, \mathcal{F}, \mu)$ be a σ -finite measure space and $1 \leq p \leq \infty$. Then $L^p(\mu)$ is reflexive (equivalently, the canonical map $J : L^p(\mu) \rightarrow (L^p(\mu))^{**}$ is surjective) if and only if

$$1 < p < \infty.$$

Equivalently,

$$(L^p(\mu))^{**} \cong L^p(\mu) \iff 1 < p < \infty.$$

Proof. Recall that a Banach space X is called *reflexive* if the canonical embedding

$$J : X \longrightarrow X^{**}, \quad J(x)(\varphi) := \varphi(x) \quad (\varphi \in X^*),$$

is surjective. It is always an isometric linear embedding.

Step 1: Duality for $1 < p < \infty$.

Let $1 < p < \infty$ and denote by q the conjugate exponent, $1/p + 1/q = 1$. A standard result (Riesz representation for L^p) says that

$$(L^p(\mu))^* \cong L^q(\mu)$$

isometrically: to each $g \in L^q(\mu)$ we associate the bounded linear functional

$$T_g(f) := \int_{\Omega} fg d\mu, \quad f \in L^p(\mu),$$

and every $\Phi \in (L^p)^*$ is of this form, with $\|\Phi\| = \|g\|_q$.

Applying the same result with p and q interchanged, we also have

$$(L^q(\mu))^* \cong L^p(\mu).$$

Hence

$$(L^p(\mu))^{**} = ((L^p(\mu))^*)^* \cong (L^q(\mu))^* \cong L^p(\mu)$$

as Banach spaces. One checks (or notes as standard) that under these identifications the isomorphism $L^p \rightarrow (L^p)^{**}$ coincides with the canonical embedding J , hence J is surjective. Therefore $L^p(\mu)$ is reflexive for $1 < p < \infty$.

Step 2: $L^1(\mu)$ is not reflexive (infinite-dimensional case).

Assume now that $L^1(\mu)$ is infinite-dimensional and reflexive. It is a standard fact that $L^1(\mu)$ is separable (for σ -finite μ). If a Banach space X is reflexive and separable, then its dual X^* is also separable (the unit ball of X^* is weak-* metrizable and compact).

For σ -finite μ we have

$$(L^1(\mu))^* \cong L^\infty(\mu).$$

However, $L^\infty(\mu)$ is *not* separable (again a standard fact whenever $L^1(\mu)$ is infinite-dimensional). This contradicts the separability of $(L^1)^*$ that would follow from reflexivity of L^1 . Hence $L^1(\mu)$ cannot be reflexive (unless it is finite-dimensional).

Therefore, in the infinite-dimensional case, $L^1(\mu)$ is not reflexive, so $(L^1(\mu))^{**} \not\cong L^1(\mu)$.

Step 3: $L^\infty(\mu)$ is not reflexive (infinite-dimensional case).

Suppose, toward a contradiction, that $L^\infty(\mu)$ is reflexive and infinite-dimensional. A general fact from Banach space theory is that if X is reflexive, then its dual X^* is also reflexive.

It is also known that $(L^\infty(\mu))^*$ contains a closed subspace isometrically isomorphic to $L^1(\mu)$ (for instance via countably additive measures absolutely continuous with respect to μ). Since closed subspaces of reflexive spaces are reflexive, this would imply that $L^1(\mu)$ is reflexive, contradicting Step 2 in the infinite-dimensional case.

Thus $L^\infty(\mu)$ is not reflexive (again, unless it happens to be finite-dimensional, which we exclude by assumption).

Conclusion.

We have shown:

- For $1 < p < \infty$, $L^p(\mu)$ is reflexive, so $(L^p(\mu))^{**} \cong L^p(\mu)$.
- For $p = 1$ and $p = \infty$, assuming $L^p(\mu)$ is infinite-dimensional, $L^p(\mu)$ is not reflexive, so the canonical map $L^p(\mu) \rightarrow (L^p(\mu))^{**}$ is not surjective.

This proves that, in the infinite-dimensional case,

$$(L^p(\mu))^{**} \cong L^p(\mu) \iff 1 < p < \infty.$$

□

Definition 9.12 [$C_c(x)$]. $C_c(x)$ = continuous functions on X with compact support. i.e. $C_c(\mathbb{N}) = \{\text{sequences } f : \mathbb{N} \rightarrow \mathbb{C} \text{ with finitely many non-zero terms}\}$

Theorem 9.13 (Rietz-Markov Theorem). *Let X be a locally compact Hausdorff space and ψ a positive linear functional on $C_c(X)$. Then there exists a unique positive Borel measure μ on X such that*

$$\psi(f) = \int_X f(x) d\mu(x), \quad \forall f \in C_c(X),$$

which has the following additional properties for some Σ containing the Borel σ -algebra on X :

- $\mu(K) < \infty$ for every compact $K \subset X$,
- **Outer regularity:** $\mu(E) = \inf\{\mu(U) : E \subset U, U \text{ open}\}$ for every Borel set $E \in \Sigma$,
- **Inner regularity:** $\mu(E) = \sup\{\mu(K) : K \subset E, K \text{ compact}\}$ whenever E is open or when E is Borel and $\mu(E) < \infty$,
- (X, Σ, μ) is a complete measure space.

As such, if all open sets in X are σ -compact then μ is a Radon measure.

9.1 $(L^\infty)^*$

To study $L^\infty(\mathbb{R})^*$, we study $L^\infty(\mathbb{N})$ first. Here $L^\infty(\mathbb{N})$ is the continuous and bounded functions on $\mathbb{N}$. The Stone-Čech compactification is given as

$$\widehat{\mathbb{N}} = \{\text{maximal ideals in } C_b(\mathbb{N})\}.$$

Then

$$\mathbb{N} \longrightarrow \widehat{\mathbb{N}}, \quad n \longmapsto \ker(e_n : C_b(\mathbb{N}) \rightarrow \mathbb{C}),$$

where e_n is the evaluation functional at n .

We can put a topology on $\widehat{\mathbb{N}}$. Then

$$L^\infty(\mathbb{N}) = C(\widehat{\mathbb{N}}).$$

so

$$L^\infty(\mathbb{N})^* = C(\widehat{\mathbb{N}})^* = \text{Space of Radon measures on } \widehat{\mathbb{N}}$$

If we try to do a similar construction for $L^\infty(\mathbb{R})$, we encounter a difficulty. For the space of bounded continuous functions we have

$$C_b(\mathbb{R}) \cong C(\beta\mathbb{R}),$$

where $\beta\mathbb{R}$ is the Stone-Čech compactification of $\mathbb{R}$. This gives a canonical compact Hausdorff space such that every function in $C_b(\mathbb{R})$ extends uniquely to $\beta\mathbb{R}$.

However, for $L^\infty(\mathbb{R})$ there is no unique choice of compact space. There exist many **compact Hausdorff** spaces K such that

$$L^\infty(\mathbb{R}) \cong C(K),$$

but these choices of K are not canonical.

Nevertheless, for any such K we always have a continuous surjection

$$\beta\mathbb{R} \twoheadrightarrow K,$$

because $C(K)$ arises as a quotient of $C(\beta\mathbb{R})$.

Using the Riesz representation theorem for compact Hausdorff spaces, this yields

$$L^\infty(\mathbb{R})^* = C(K)^* \cong \text{the space of Radon measures on } K.$$

Thus the dual of $L^\infty(\mathbb{R})$ can be described as the space of Radon measures on some compact Hausdorff space, but this space is not unique. This explains why $L^\infty(\mathbb{R})^*$ does not have a canonical description analogous to the dual of L^p for $1 < p < \infty$.

Theorem 9.14 (Riesz Representation Theorem (Compact Hausdorff)). *Let K be a compact Hausdorff space. For every bounded linear functional $T : C(K) \rightarrow \mathbb{C}$, there exists a unique finite Radon measure μ on K such that*

$$T(f) = \int_K f \, d\mu, \quad \forall f \in C(K),$$

and

$$\|T\| = |\mu|(K).$$

Thus

$$C(K)^* \cong \mathcal{M}(K),$$

where $\mathcal{M}(K)$ is the Banach space of finite Radon measures on K .

Definition 9.15 [Regular Measure]. A Borel measure μ is **outer regular** (That is, can be approximate by open sets from outside of the set) on $E \in \text{Bor}(X)$ if

$$\mu(E) = \inf\{\mu(U) : E \subset U \text{ open}\},$$

and **inner regular** (That is, can be approximated by compact sets from inside) on E if

$$\mu(E) = \sup\{\mu(K) : E \supset K \text{ compact}\}.$$

We call μ a *regular measure* if it is both inner and outer regular on all Borel sets. A Borel measure μ is a *Radon measure* if $\mu(K) < \infty$ for all compact sets K , it is outer regular on all Borel sets, and inner regular on open sets.

10 Final Review Recap

10.1 Derivatives

Definition 10.1 [Weak Derivative]. For a function f , we define its derivative by:

$$\int f\varphi' = - \int f'\varphi$$

In general, for $f \in L^1$,

$$\int f' \psi = 0 \quad \forall \psi \in C^1([0, 1]), \quad \psi(0) = \psi(1) = 0$$

implies

$$f = \text{constant} \quad \text{a.e.}$$

Proof. Consider

$$f_\varepsilon(x) = \int_0^1 \rho_\varepsilon(x - y) f(y) dy.$$

Then

$$f'_\varepsilon(x) = \int_0^1 \rho'_\varepsilon(x - y) f(y) dy = 0, \quad x \in (2\varepsilon, 2(1 - \varepsilon)).$$

Hence

$$f_\varepsilon(x) = c_\varepsilon \quad \text{on } (\varepsilon, 1 - \varepsilon).$$

Note that $f_\varepsilon \rightarrow f$ in L^1 , and therefore

$$f_{\varepsilon_i} \rightarrow f \quad \text{a.e.}$$

Thus

$$f = \text{constant} \quad \text{a.e.} \quad !$$

□

10.1.1 Cantor-Lebesgue

We claim that the derivative of Cantor-Lebesgue function does not exist and hence it's not a contradiction to our definition.

Proof. Suppose, for a contradiction, that f'_{CL} exist, then it's easy to see that $f'_{CL} = 0$ a.e. Then by our definition of derivative, we know that

$$\int f' \varphi = \int f \varphi' = 0,$$

for all $\varphi \in C_c$.

Since this works for all φ , we can choose a φ such that φ is increasing on the left side but decreasing on the right side. That is, $\varphi' > 0$ on the left side and $\varphi' < 0$ on the right side.

Notice that on the left side, $f > 0$ is small and $\varphi' > 0$, so $\int_{left} f \varphi'$ is a small positive value. On the other hand, $f > 0$ is large and $\varphi' < 0$ on the right side, thus $\int_{right} f \varphi' < 0$ has a large negative value. Therefore, $\int f \varphi' < 0$. This is a contradiction to $\int f \varphi' = 0$. Thus, f_{CL} does not exist.

□

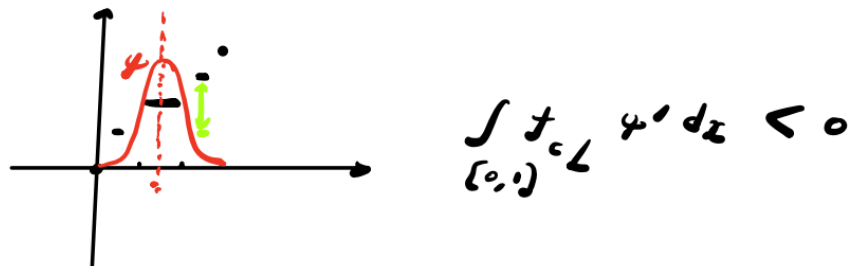


Figure 6: Construct φ

Example 10.2 [Derivative of Cantor-Lebesgue]. Compute another example:

Let $\varphi = x^2 - x$, then $\varphi' = 2x - 1$. We claim that $\int f_{CL} = \frac{1}{2}$, here's the proof:

Notice that

$$f_{\varepsilon}(x) + f_{\varepsilon}(1-x) = 1.$$

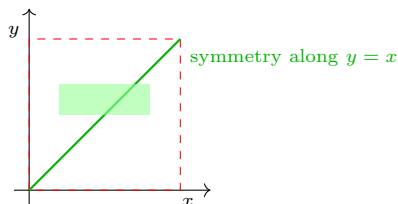
Therefore, integrating from 0 to 1,

$$2 \int_0^1 f_{\varepsilon}(x) dx = \int_0^1 f_{\varepsilon}(x) dx + \int_0^1 f_{\varepsilon}(1-x) dx = \int_0^1 (f_{\varepsilon}(x) + f_{\varepsilon}(1-x)) dx = \int_0^1 1 dx = 1.$$

Hence

$$\int_0^1 f_{\varepsilon}(x) dx = \frac{1}{2}.$$

where we use f_{ε} to denote f_{CL} .



Let us compute now

$$\int_0^1 x f_{CL}(x) dx.$$

Note:

$$x \in [0, \frac{1}{3}] \implies f_{CL}(x) = \frac{1}{2} f_{CL}(3x).$$

$$x \in [\frac{1}{3}, \frac{2}{3}] \implies f_{CL}(x) = \frac{1}{2}.$$

$$x \in [\frac{2}{3}, 1] \quad \implies \quad f_{CL}(x) = \frac{1}{2} + \frac{1}{2} f_{CL}(3x - 2).$$

Indeed, $f_{CL} : [0, \frac{1}{3}] \rightarrow [0, 1]$ for all $x \in [0, \frac{1}{3}]$ and because of symmetry of Cantor-Lebesgue function, the first interval maps to $\frac{1}{2}$.

Now we can split the integral into three parts, let's consider $x \in [0, \frac{1}{3}]$ first:
Let

$$S := \int_0^1 x f_{CL}(x) dx$$

$$A := \int_0^{\frac{1}{3}} x f_{CL}(x) dx = \frac{1}{3} \int_0^{\frac{1}{3}} \frac{3x}{6} f_{CL}(3x) d(3x)$$

Let $x = 3x$, then

$$A := \frac{1}{18} \int_0^1 x f_{CL}(x) dx = \frac{1}{18} S$$

Similar, we can show that $B := \int_{\frac{1}{3}}^{\frac{2}{3}} x f_{CL}(x) dx = \frac{1}{12}$, and $C := \int_{\frac{2}{3}}^1 x f_{CL}(x) dx = \frac{5}{36} + \frac{1}{18}(S + 1)$. Combine all three, we can solve for $S = \frac{5}{16}$.

Thus, $\int_0^1 f_{CL}(x)(2x - 1) dx = \frac{1}{8} \neq 0$. Therefore, f_{CL} does not exist.

10.2 Note

For continuous, non-decreasing functions f, g ,

$$d(fg) = gdf + fdg$$

so

$$\int_{(a,b]} gdf = f(b)g(b) - f(a)g(a)$$